

Notes on Master Studies

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Abstract

“En remontant chez moi pour y passer la soirée à travailler de mon mieux, je me disais que le monde n’est pas construit pour l’équilibre. Le monde est désordre. L’équilibre n’est pas la règle, c’est l’exception.”
G.Duhamel, Maitres, 1937

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Chapter 1

Random Matrix Theory

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix. This is the case for the invariant ensembles. With the help of some mathematical tools such as orthogonal polynomials one can express the equilibrium measures for the eigenvalues asymptotically.

1.1 Random Matrices

This is based on the [1]

An ensemble can be thought as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. More concretely, in the context of Random Matrix Theory

Definition 1.1.1. A random matrix ensemble $E = (S, P)$ is a set of matrices of fixed size equipped with a probability density function (p.d.f) $P : S \rightarrow [0, \infty)$. If we say that a matrix X is drawn from the random ensemble E , we are saying that it is a random variable assuming values in S with p.d.f. P .

A commonplace thing to do when defining ensembles is to prescribe the p.d.f's $P_{i,j}$ on the independent entries $X_{i,j}$ of a representative matrix $X \in E$. In such cases, we say that arXiv:2301.11428v1 [math-ph] 26 Jan 2023

$$P(X) = \prod_{i,j | X_{i,j} \text{ independent}} P_{i,j}(X_{i,j}).$$

With this definition we can already define many ensembles of interest. For many, the set of matrices is described by applying constraints to the set of real matrices $M_{m,n}(\mathbb{R})$, the set of complex matrices $M_{m,n}(\mathbb{C})$ or the set of quaternionic matrices $M_{m,n}(\mathbb{H})$. Then, oftentimes, S is simply the set of symmetric, Hermitian or skew-symmetric matrices. When it comes to determining the p.d.f P on S , there are many options that have been worked on. A simple example is that of *Bernoulli matrices* B whose entries $B_{i,j}$ are all independently and identically distributed Bernoulli random variables ± 1 : $S = M_{n,n}(\mathbb{R})$ and

$$P_{i,j}(B_{i,j}) = \frac{1}{2} [\delta(B_{i,j} - 1) + \delta(B_{i,j} + 1)].$$

If we instead require B to be symmetric, its diagonal entries to be zero, and its independent entries have p.d.f.

$$P_{i,j}(B_{i,j}) = \frac{c}{n} \delta(B_{i,j} - 1) + \left(1 - \frac{c}{n} \delta(B_{i,j})\right),$$

then B belongs to the ensemble of *Erdos-Renyi*.

Definition 1.1.2. (*Ginibre Ensemble*). The $m \times n$ real, complex, and quaternionic Ginibre ensembles are respectively $M_{m,n}(\mathbb{R})$, $M_{m,n}(\mathbb{C})$, $M_{m,n}(\mathbb{H})$ with p.d.f

$$p^{(Gin)}(G) = \pi^{-mn \frac{\beta}{2}} \exp(-\text{Tr}(G^\dagger G)) = \prod_{i=1}^m \prod_{j=1}^n \pi^{\frac{\beta}{2}} \exp(-|G_{i,j}|^2),$$

defined in respect to the Lebesgue measure

$$dG = \prod_{i=1}^m \prod_{j=1}^n \prod_{s=1}^{\beta} dG_{i,j}^{(s)}.$$

The coefficient β is called the *Dyson index* and counts the generic number of real components of each matrix entry $G_{i,j}$, so that $\beta = 1, 2, 4$ refers to real, complex and quaternionic entries. These are the Ginibre orthogonal, unitary and symplectic ensembles.

For most applications there are three ideologies that stand out. The first one is to assume little on the system and focus on universal results. This is the *Wigner matrix ensembles*: Let X be drawn from either the real symmetric, complex hermitian or quaternionic skew-symmetric $n \times n$ matrix ensemble. Then its entries $X_{i,j}$ are independently distributed with mean zero and bounded moments. Also, its diagonal entries are equally distributed as are its upper triangular entries. Many results hold for independently and identically distributed centered random variables.

The second ideology is assuming equal probability on each state of the system - this is a common assumption in thermodynamics. So this approach is to assume that every matrix is equally likely.

Observation 1.1.3. We have focused on random matrix ensembles distributed according to p.d.f.s P defined with respect to the flat Lebesgue measure induced by the canonical embedding of S into Euclidean space. Note here, which was not obvious for me, that the measures ν we define are given in terms of the p.d.f.s P and the Lebesgue measure μ by Radon-Nikod in

$$\nu(A) = \int_A P d\mu,$$

or more simply, $d\nu = P d\mu$.

Theorem 1.1.4. (*Haar, compact case*). Given a compact group S , let $U \in S$ be a generic variable. There exists a unique (up to normalization measure) $d\mu(U)$ on S , called the **Haar measure** such that

$$\int_S d\mu(U) < \infty$$

and for any fixed $U_0 \in S$,

$$d\mu(U_0 U) = d\mu(U) = d\mu(U U_0),$$

that is, finite for compact sets and invariant by left and right actions. If $d\mu(U)$ integrates to unity in S it is referred as the **Haar probability measure**.

Definition 1.1.5. The $n \times n$ circular unitary ensemble (CUE) is the unitary group $U(n)$ with its Haar measure. If $U \in CUE(n)$, then the symmetric unitary matrix $U^T U$ represents the $n \times n$ circular orthogonal ensemble (COE); we have $S \cong U(n)/O(n)$. Note that if $U \in U(n)$ and $O \in O(n)$, we have that $U^T U = (OU)^T (OU)$, that is, U and OU are in the same co-class.

The third ideology is that of working with algebraic combinations of matrices whose entries are independently and identically distributed.

1.1.1 Some quantities

Assume a matrix is such that is able to be determined by eigenvalues and eigenvectors. While eigenvector may be of interested in some cases, eigenvalue statistics have been considerably more explored. Hence, in addition to the p.d.fs $P(X)$ of say $n \times n$ random matrices X , we are also interested in the eigenvalue j.p.d.f $p(\lambda_1, \lambda_2, \dots, \lambda_n)$ of the eigenvalues $\{\lambda_i\}_{i=1}^n$ of X . The eigenvalue j.p.d.f is defined to be such that

$$\int_{[a_1, b_1] \times \dots \times [a_n, b_n]} p(\lambda_1, \lambda_2, \dots, \lambda_n) d\lambda_1 \dots d\lambda_n$$

is the probability that $a_i < \lambda_i < b_i$ for every i .

Example 1.1.6. The eigenvalue j.p.d.f of the $n \times n$ circular ensembles is

$$p^{(C)}(e^{i\theta_1}, \dots, e^{i\theta_n}, \beta) = \frac{1}{Z_{n,\beta}^{(C)}} |\Delta_N(e^{i\theta})|^\beta$$

where $Z_{n,\beta}^{(C)}$ is a normalization constant and

$$\Delta_N(\lambda) := \det \left[\lambda_i^{j-1} \right]_{i,j=1}^n = \prod_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)$$

is the Vandermonde determinant, and $\beta = 1, 2, 4$ correspond to the COE, CUE and CSE, respectively.

Related do the eigenvalue j.p.d.f is the univariate eigenvalue density defined by

$$p(\lambda_1, n) := n \int_{\mathbb{R}^{N-1}} p(\lambda_1, \lambda_2, \dots, \lambda_n) d\lambda_1 \dots d\lambda_n,$$

which is normalized such that the expect number of eigenvalues lying between a and b is given by $\int p(\lambda) d\lambda$. Note that this is not a p.d.f is $n \neq 1$. Consider now the linear statistic of the eigenvalue density given by

$$p(\lambda) = \sum_{i=1}^n \langle \delta(\lambda - \lambda_i) \rangle$$

where the brackets are understood as acting on an operator as

$$\langle \mathcal{O} \rangle = \int_{\mathbb{R}^n} \mathcal{O} p(\lambda_1, \dots, \lambda_n) d\lambda_1 \dots d\lambda_n.$$

Many times, eigenvalues density are fully characterized by their *spectral moments*

$$m_k := \int_{\mathbb{R}} \lambda^k p(\lambda) d\lambda, \quad k \in \mathbb{N}.$$

Indeed, if we are interested in a linear statistic $\sum_{i=1}^n f(\lambda_i)$ of the eigenvalues such that $f(\lambda)$ is analytic at zero with Taylor expansion $f(\lambda) = \sum_{k=0}^{\infty} f_k \lambda^k$, we can compute its expectation to be

$$\left\langle \sum_{i=1}^n f(\lambda_i) \right\rangle = \int_{\mathbb{R}} f(\lambda) p(\lambda) d\lambda = \sum_{k=0}^{\infty} f_k m_k$$

The spectral moments can additionally be expressed as an average with respect to the eigenvalue j.p.d.f $p(\lambda_1, \lambda_2, \dots, \lambda_n)$ and the p.d.f $P(X)$ of the random matrix X according to

$$m_k = \left\langle \sum_{i=1}^n \lambda_i^k \right\rangle = \langle \text{Tr}(X^k) \rangle_{P(X)}$$

where we define

$$\langle \mathcal{O} \rangle_{P(X)} := \int_S \{ \mathcal{O} P(X) \} dX$$

Moment Generating Function

We can also define an useful tool for studying the spectral moments: The *Moment Generating Function* given by

$$W_1(x) := \sum_{k=0}^{\infty} \frac{m_k}{x^{k+1}},$$

which is also known as the *Resolvent* due to it being equal to the *Stieltjes Transform* of the eigenvalue density

$$W_1(x) = \int_{\text{supp} p} \frac{p(\lambda)}{x - \lambda} d\lambda, \quad \text{for } x \in \mathbb{C} \setminus \text{supp} p.$$

Obtaining a closed form expression for the resolvent allows one to use the *Sokhotski-Plemelj* inversion formula to recover the eigenvalue density

$$p(\lambda) = \lim_{\epsilon \rightarrow 0} \frac{W_1(\lambda - i\epsilon) - W_1(\lambda + i\epsilon)}{2\pi i}$$

as long as $p(\lambda)$ is known to be continuous on its support. As we did before, we can re-express the resolvent as an ensemble average w.r.t to the eigenvalue j.p.d.f $p(\lambda_1, \lambda_2, \dots, \lambda_n)$ and the p.d.f $P(X)$ on S :

$$W_1(x) = \left\langle \sum_{i=1}^n \frac{1}{x - \lambda_i} \right\rangle = \left\langle \text{Tr} \frac{1}{x - X} \right\rangle_{P(X)}$$

These three expressions facilitate the dealing of the eigenvalue densities. In fact, dealing with the resolvent will be many times preferred as it presents an asymptotic expansion the densities does not.

Theorem 1.1.7. (*Borot-Guionnet*) Assuming certain hypothesis on the eigenvalues j.p.d.f. $p(\lambda_1, \lambda_2, \dots, \lambda_N)$, and the resolvent $W_1(x)$ and, more generally, the undefined connected n -point correlator $W_n(x_1, x_2, \dots, x_n)$, have large N expansions of the form

$$W_n(x_1, x_2, \dots, x_n, N) = N^{2-n} \sum_{l=0}^{\infty} \frac{W_n^l(x_1, x_2, \dots, x_n)}{N^l},$$

where the correlator expansion coefficients $W_n^l(x_1, x_2, \dots, x_n)$ have no dependence on n .

Some of the conditions of the theorem above guide us to rescale the eigenvalues of certain ensembles so that they are contained in a compact connected set in the large limit $N \rightarrow \infty$ regime. In these cases, we study the *global scaled eigenvalue density* $\tilde{p}(\lambda)$ and the *scaled resolvent* $\tilde{W}_1(x)$ given by

$$\tilde{p}(\lambda) := \frac{c_N}{N} p(c_N \lambda)$$

$$\tilde{W}_1(x) := c_N W_1(c_N x) = N \int_{\text{supp } \tilde{p}} \frac{\tilde{p}(\lambda)}{x - \lambda} d\lambda$$

where c_N is a scaling parameter that differs from ensemble to ensemble. Now, assuming the theorem holds, one has that $\tilde{W}_1 = \sum_{l=1}^{\infty} W_1^l(x) N^{1-l}$. Using the Sokhotski-Plemelj inversion formula we define

$$p^l(\lambda) := \lim_{\epsilon \rightarrow 0} \frac{W_1^l(\lambda - i\epsilon) - W_1^l(\lambda + i\epsilon)}{2\pi i}.$$

Then, $\tilde{p}(\lambda)$ and $\sum_{l=1}^{\infty} p^l(\lambda) N^{1-l}$ do not agree at finite N , as one might expect. Instead, the latter sum is referred to as the *smoothed eigenvalue density*, as it lacks the oscillatory terms that are known to be present in the true scaled eigenvalue density $\tilde{p}(\lambda)$.

Classical Ensembles

The denomination Classical Ensemble usually denotes the Gaussian, Laguerre and Jacobi ensembles because of their future discussed relation to the orthogonal polynomials.

Definition 1.1.8. We define the following classical ensembles

- Let G be drawn from the $n \times n$ real, complex or quaternionic Ginibre Ensemble. Then, the random matrix $H = \frac{1}{2}(G^\dagger + G)$ represents the corresponding Gaussian ensemble (also known as Hermite).
- Let G be drawn from the $n \times m$ real, complex or quaternionic Ginibre Ensemble. Then, the random matrix $W = (G^\dagger G)$ represents the corresponding (m, n) Laguerre ensemble (also known as Wishart).
- We say Y an element of the (m_1, m_2, n) real, complex and quaternionic Jacobi Ensemble if it is of the form $Y = W_1(W_1 + W_2)^{-1}$ where the W_i belong to the corresponding (m_i, n) Laguerre ensembles with $m_i \geq n$.

Note that all ensembles presented are self-adjoint and therefore have all n eigenvalues. Gaussian ensembles are also examples of Wigner Matrices.

Proposition 1.1.9. Let $k := \beta/2$, $a = k(m_1 - n + 1) - 1$ and $b = k(m_2 - n + 1) - 1$. From the p.d.f. of the Ginibre ensembles, we can change variables to obtain the p.d.f.s of the $n \times n$ Gaussian, (m_1, n) Laguerre, and (m_1, m_2, n) Jacobi ensembles, respectively

$$p^{(G)}(H) = \frac{1}{Z_{n,\beta}^{(G)}} \exp(-\text{Tr}(H^2)) \quad (1.1.1)$$

$$p^{(L)}(W) = \frac{1}{Z_{n,a}^{(L)}} (\det(W))^a \exp(-\text{Tr}(W)) \quad (1.1.2)$$

$$p^{(J)}(Y) = \frac{1}{Z_{n,a,b}^{(J)}} (\det(Y))^a (\det(I_n - Y)) \quad (1.1.3)$$

With $\beta = 1, 2, 4$ corresponding to the choice of $F = \mathbb{R}, \mathbb{C}, \mathbb{H}$, the random matrix ensembles introduced in the definition are defined introducing these p.d.fs to the matrix sets

$$\begin{aligned} S^{(G)} &= \{H \in M_{n,n}(F) | H = H^\dagger\} \\ S^{(L)} &= \{W \in S^{(G)} | W \text{ positive definite}\} \\ S^{(J)} &= \{Y \in S^{(L)} | I_n - Y \text{ positive definite}\} \end{aligned} \quad (1.1.4)$$

The contemporary definition of classical ensembles introduces another ensemble into the family.

Definition 1.1.10. A random matrix ensemble is said to be a classical matrix ensemble if its eigenvalue j.p.d.f is of the form

$$p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta, \quad \beta = 1, 2, 4,$$

where $\mathcal{N}_{n,\beta}^{(w)}$ is the normalization constant, $\Delta_n(\lambda)$ is the Vandermonde determinant and the weight function $w(\lambda)$ is such that

$$\frac{d}{d\lambda} \log(w(\lambda)) = \frac{w'(\lambda)}{w(\lambda)} = \frac{f(\lambda)}{g(\lambda)}$$

with f and g polynomials of degree at most one and two, respectively.

Up to fractional linear transformations $\lambda \mapsto (a_{11}\lambda + a_{12})/(a_{21}\lambda + a_{22})$ with $a_{ij} \in GL_2(\mathbb{R})$, there are precisely 4 weight function that satisfies the condition.

$$w(\lambda) = \begin{cases} e^{-\lambda^2} & \text{Gaussian,} \\ \lambda^a e^{-\lambda} \chi_{\lambda>0} & \text{Laguerre,} \\ \lambda^a (1-\lambda)^b \chi_{0<\lambda<1} & \text{Jacobi,} \\ (1-i\lambda)^\eta (1+i\lambda)^{\bar{\eta}} & \text{generalized Cauchy.} \end{cases} \quad (1.1.5)$$

where χ is the indicator function.

Definition 1.1.11. Taking $k = \beta/2$ and $c_n = \sqrt{kn}, kn, 1$ respectively for the Gaussian, Laguerre and Jacobi ensembles. The global scaled eigenvalue densities of the classical matrix ensembles are

$$\begin{aligned} \tilde{p}^{(G)}(\lambda) &= \sqrt{\frac{k}{n}} p^{(G)}(\sqrt{kn}\lambda) \\ \tilde{p}^{(L)}(\lambda) &= k p^{(L)}(kn\lambda) \\ \tilde{p}^{(J)}(\lambda) &= \frac{1}{n} p^{(J)}(\lambda) \\ \tilde{p}^{(Cy)}(\lambda) &= \frac{1}{n} p^{(J)}(\lambda) \end{aligned} \quad (1.1.6)$$

1.2 Invariant Measure

Given a group $\mathcal{G}$ and a group action $\mathcal{A} : \mathcal{G} \times S \rightarrow S$, a random matrix ensemble $E = (S, P)$ is said to be $\mathcal{A}$ -invariant if the probability measure $dP(X) := P(X)dX$ satisfies

$$dP(\mathcal{A}(g, X)) = dP(X)$$

for every $g \in \mathcal{G}$. The term *invariant matrix ensemble* usually refers to invariance under the mapping $X \mapsto gXg^{-1}$ for g drawn from the appropriate compact group $\mathcal{G}$; when a matrix ensemble is considered to be irreducible and its underlying set of matrix is $S \subseteq M_{n,n}(\mathbb{R}), M_{n,n}(\mathbb{C}), M_{n,n}(\mathbb{H})$, the group $\mathcal{G}$ must be one of $O(n), U(n), Sp(2n)$. Then, we say the ensemble is orthogonal, unitary or symplectic.

We can also determine the probability measure $dP(X)$ to be invariant under the adjoint action of a group $\mathcal{G}$ if we establish that both the p.d.f $P(X)$ and the Lebesgue measure dX are invariant under this action. We can say directly using 1.1.2 and 1.1.9 that the p.d.f.s for Ginibre, Gaussian, Laguerre and Jacobi are invariant under the appropriate adjoint actions.

Proof. (Ginibre, Gaussian, Laguerre and Jacobi are invariant under the appropriate adjoint actions.) $\square$

The invariance of the Lebesgue measure is more involved and depends on whether or not the associated matrix set S consists of purely of self-adjoint matrices. We start with the case of the Gaussian, Laguerre and Jacobi ensembles.

Proposition 1.2.1. *Given an $n \times n$ real ($\beta = 1$), complex ($\beta = 2$) or quaternionic ($\beta = 4$) self-adjoint matrix variable X , diagonalize it as $X = U\Lambda U^\dagger$ where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is a diagonal matrix of eigenvalues, and U is corresponding matrix of eigenvectors; the spectral theorem dictates that U is an element of $O(n)$ in the real case, $U(n)$ in the complex case, and $Sp(2n)$ in the quaternionic case. To make the mapping $X \mapsto U\Lambda U^\dagger$ bijective, we insist that the eigenvalues be listed in increasing order and the first row of U to be non-negative real. The canonical Lebesgue measure*

$$dX = \prod_{i=1}^n dX_{ii} \prod_{1 \leq j < k \leq n} \prod_{s=1}^{\beta} dX_{jk}^s$$

on the set of adjoint matrices decomposes as

$$dX = |\Delta_n|^\beta d\Lambda d\mu_{Haar}(U)$$

where $d\Lambda = \prod_{i=1}^n d\lambda_i$ is the Lebesgue measure on the Weyl Chamber $\lambda_1 < \dots < \lambda_n \subset \mathbb{R}^n$ and

$$d\mu_{Haar}(U) = \bigwedge_{1 \leq l < k \leq n} \bigwedge_{s=1}^{\beta} (U^\dagger [dU_{ij}]_{i,j=1}^n)_{lk}^{(s)}$$

is the unnormalized Haar measure on the quotient space $O(n)/O(1)^n$ - $\beta = 1$, $U(n)/U(1)^n$ - $\beta = 2$, or $Sp(2n)/Sp(2)^n$ - $\beta = 4$, given by the exterior product of the independent real components of the entries of the matrix of differentials $U^\dagger [dU_{ij}]_{i,j=1}^n$.

At this point, one can check (albeit with some non-trivial computations) that the Lebesgue measure is indeed invariant in these cases for the appropriate group action.

The situation concerning the $n \times n$ Ginibre ensemble is more complicated. In the 1965 work, Ginibre extended the proposition above and its arguments starting with the diagonalization formula $G = P\Lambda P^{-1}$ - note that, with Λ diagonal of eigenvalues and P of eigenvectors, this formula is valid with probability one since the subset of matrices with repeated eigenvalues has measure zero. However, P , diagonalizing matrix, is not necessarily orthogonal, unitary, nor symplectic. Nonetheless, QR decomposition can be used to write $P = UT$ where U is as before and T is an upper triangular matrix with real positive diagonal entries. Then,

$$dG = J(\lambda) d\Lambda d\mu_{Tri}(T) d\mu_{Haar}(U)$$

where $d\Lambda$ is the Lebesgue measure on the space housing the eigenvalues, $d\mu_{Haar}(U)$ is the Haar measure defined before, $J(\lambda)$ is a Jacobian consisting of products of Vandermonde determinants involving the eigenvalues, and $\mu_{Tri}(T)$ is some measure dependent on the entries of T . From here, it follows that the Lebesgue measure on G is invariant under the desired conjugation action.

Eigenvalues j.p.d.fs

To obtain the eigenvalue j.p.d.f $p(\lambda_1, \lambda_2, \dots, \lambda_n)$ from the probability measure $dP(X)$, one changes variable to the eigenvalues and a set of auxiliary variables, then integrates over the latter. For the self-adjoint cases, these auxiliary variables were the quotient spaces of orthogonal, unitary and symplectic matrices. For Ginibre, is the upper triangular entries.

Leaving the details of Ginibre behind, we focus on the Gaussian, Laguerre and Jacobi ensembles. Set $P(X)$ the p.d.f. for one of the ensembles, fix $(\beta, \mathcal{G}(n)) = (1, O(n)), (2, U(n)), (4, Sp(2n))$. The propositions cited implies that

$$\int_{(G)(n)/\mathcal{G}(1)^n} dP(X) = P(\Lambda) |\Delta_n(\lambda)|^\beta d\Lambda \int_{(G)(n)/\mathcal{G}(1)^n} d\mu_{Haar}(U)$$

As the eigenvalue and eigenvector statistics are decoupled, one reads off the eigenvalue j.p.d.f is thus given by

$$p(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{\text{Vol}((G)(n)/\mathcal{G}(1)^n)}{N!} P(\Lambda) |\Delta_n(\lambda)|^\beta,$$

where, of course,

$$\text{Vol}((G)(n)/\mathcal{G}(1)^n) = \int_{(G)(n)/\mathcal{G}(1)^n} d\mu_{Haar}(U)$$

Now, with $w(\lambda)$ the appropriate classical weight, we can check that

$$P(\Lambda) = \prod_{i=1}^n w(\lambda_i),$$

which confirms the consistence with the definition of the eigenvalue j.p.d.f on the classical ensembles. Furthermore, note that

$$Z_n = \frac{\text{Vol}((G)(n)/\mathcal{G}(1)^n)}{N!} N_{n,\beta}$$

as long as the volume is known.

Invariant ensembles of random matrices are probability spaces which are invariant under conjugation of the appropriate matrix group - these ensembles only depend on their eigenvalue distributions while their eigenvectors become independent and are Haar distributed. To properly understand this concept, let's make precise the definition of three infamous ensembles: Unitary, Orthogonal and Symplectic ensembles. Other ensembles many times can be of interest, but their theory may be less straightforward.

1. *Orthogonal Ensembles* consist of $n \times n$ real symmetric matrices $\hat{M} = \hat{M}^T$ together with a probability distribution that is invariant under orthogonal conjugation $\hat{M} \mapsto \hat{O} \hat{M} \hat{O}^T$, $\hat{O}$ orthogonal, i.e., $\hat{O} \hat{O}^T = \hat{O}^T \hat{O} = \hat{I}$.
2. *Unitary Ensembles* consists of $n \times n$ Hermitian matrices $\hat{M} = \hat{M}^*$ together with a probability distribution that is invariant under unitary conjugation $\hat{M} \mapsto \hat{U} \hat{M} \hat{U}^*$, $\hat{U}$ unitary, i.e., $\hat{U} \hat{U}^* = \hat{U}^* \hat{U} = \hat{I}$.
3. *Symplectic Ensembles* consists of $2m \times 2m$ Hermitian self-dual matrices $\hat{M} = \hat{M}^* = \hat{J} \hat{M}^T \hat{J}^T$ where $\hat{J} = \text{diag}(\sigma, \dots, \sigma)$ with

$$\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

together with a probability distribution that is invariant under unitary-symplectic conjugation $\hat{M} \mapsto \hat{U} \hat{M} \hat{U}$, with $\hat{U}$ unitary and symplectic, i.e. $\hat{U} \hat{U}^* = \hat{U}^* \hat{U} = \hat{I}$ and $\hat{U} \hat{J} \hat{U}^T = \hat{J}$.

Examples of such invariant ensembles are the Gaussian Orthogonal, Unitary and Symplectic ensembles - *GOE*, *GUE*, *GSE*. These take gaussian distributed and independent entries. We may also choose to include the Ginibre Ensembles in the types of invariance we will work with. The three types of Ginibre Ensembles: Ginibre Orthogonal, Unitary and Symplectic are matrices with independent, identically distributed standard complex Gaussian entries. In this case, the eigenvalues are distributed in the complex plane. For invariant ensembles, we know that $p(\hat{M}) = p(\hat{M}')$ if $\hat{M} \equiv \hat{M}'$ by the natural equivalence relation given by the symmetry by, say, $\hat{U}$. Therefore

$$\begin{aligned} p(\hat{M}) d\hat{M} &= p(M_{1,1}, \dots, M_{n,n}) \prod_{i,j} dM_{i,j} \\ &= p\left(M_{1,1}(\hat{M}', \hat{U}), \dots, M_{n,n}(\hat{M}', \hat{U}) | J(\hat{M} \mapsto \{\hat{M}', \hat{U}\})\right) d\hat{M}' d\hat{U} \\ &\stackrel{\text{invariance}}{=} p(M_{1,1}(\hat{M}'), \dots, M_{n,n}(\hat{M}') | J(\hat{M} \mapsto \{\hat{M}'\})) d\hat{M}' \end{aligned} \tag{1.2.1}$$

Therefore we can write

$$\begin{aligned} p(\hat{M}') d\hat{M}' &= d\hat{M}' \int_{\mathbb{V}_N} p(M_{1,1}(\hat{M}'), \dots, M_{n,n}(\hat{M}') | J(\hat{M} \mapsto \{\hat{M}'\})) d\hat{U} \\ &= p(M_{1,1}(\hat{M}'), \dots, M_{n,n}(\hat{M}') | J(\hat{M} \mapsto \{\hat{M}'\})) \text{Vol}(\mathbb{V}_N) d\hat{M}' \end{aligned} \quad (1.2.2)$$

We can now dissect term by term what we need for this expression

1. The *Jacobian* is given by

$$\prod_{i \leq j} |x_j - x_i|^\beta$$

where β depends on the choice of ensemble - 1 for Orthogonal, 2 for Unitary and 4 for Symplectic.

2. The expression for

$$p(M_{1,1}(\hat{M}'), \dots, M_{n,n}(\hat{M}'))$$

must be given by $p(M) = \phi(\text{Tr}(\hat{M}), \dots, \text{Tr}(\hat{M}^n))$.

1.2.1 Wigner Matrices

Gaussian ensembles are invariant ensembles inside another important class: Wigner Matrices. To construct the general form of such matrices take the matrix $\hat{Y} = \{Y_n\}_{i,j}$ with $Y_{i,j} = Y_{j,i}^*$ and such that

1. $\{Y_{i,j}\}_{1 \leq i \leq j \leq n}$ are independent;
2. $\{Y_{i,j}\}$ are identically distributed. $\{Y_{i,i}\}$ are identically distributed;
3. $\mathbb{E}(Y_{i,j}^k) < \infty$, i.e, $r_k = \max(\mathbb{E}(Y_{i,j}^k), \mathbb{E}(Y_{i,j}^k)) < \infty$.

Definition 1.2.2. Let $\{Y_n\}_{i,j}$ be as before and set $r_2 < \infty$ with $\mathbb{E}(Y_{1,2}^2) > 0$. Then $\hat{X}_n = n^{-\frac{1}{2}} \hat{Y}_n$ is what we call a *Wigner Matrix*.

The assumptions taken are not as unnatural as they may seem at first. To properly understand this definition, note that if $\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2$ are the ordered eigenvalues of $\{Y_n\}_{i,j}$, we can estimate that

$$\lambda_1^2 + \dots + \lambda_n^2 \leq n \max_{j=1}^n (\lambda_j^2) = n \max(\lambda_1^2, \lambda_n^2)$$

therefore, if we want the eigenvalues to be limited when we take large matrices we should impose that $1/n \cdot (\lambda_1^2 + \dots + \lambda_n^2)$ must be limited. This is the Hilbert-Schmidt or Frobenius norm for these normal matrices:

$$\|\hat{Y}\|_{HS} = \frac{1}{n} \sum_{i=1}^n \lambda_i^2 = \frac{1}{n} \sum_{i,j} Y_{i,j}^2 = \frac{1}{n} \text{Tr}(\hat{Y}^2)$$

then if we only take $\hat{X}_n = \alpha_n \hat{Y}_n$

$$\begin{aligned} \mathbb{E}(\hat{X}_n) &= \frac{\alpha_n^2}{n} \sum_{1 \leq i,j \leq n} \mathbb{E}(\hat{Y}_n) \\ &= \alpha_n \mathbb{E}(Y_{1,1}^2) + (n-1) \alpha_n^2 \mathbb{E}(Y_{1,2}^2) \end{aligned} \quad (1.2.3)$$

them it is clear that there are two cases

1. $\mathbb{E}(Y_{1,2}^2) = 0$: This is the non-interesting case where $\alpha_n \rightarrow 1$
2. $\mathbb{E}(Y_{1,2}^2) > 0$: Here, we must take $\alpha_n \rightarrow n^{-\frac{1}{2}}$

Now, it is likely more clear the motivation of the definition: limit the eigenvalues to a compact region.

1.2.2 Gaussian Ensembles

Let's make it explicit the probability distribution on Gaussian Ensembles. We define $\hat{M} = n^{-\frac{1}{2}} \hat{M}' \in \mathcal{M}_{\mathbb{S}}(n)$ to be of the Gaussian Ensemble if the entries of $\hat{M}'$ are independent and distributed as follows:

$$M'_{i,j} \sim \begin{cases} \mathcal{N}_{\mathbb{S}}(0, 1/2) & \text{for } i \neq j, \\ \mathcal{N}_{\mathbb{S}}(0, 1) & \text{for } i = j, \end{cases} \quad \text{if } i \leq j.$$

where $\mathbb{S}$ is such as the real line, complex plane or quaternion space. Note that this turns $\hat{M}$ into a Wigner Matrix. Now, take for example $\mathbb{S} = \mathbb{R}$ and

$$p(\hat{M}) = \prod_{i=1}^n \frac{\exp\left(-\frac{M_{i,i}^2}{2}\right)}{\sqrt{2\pi}} \prod_{i < j} \frac{\exp\left(-M_{i,i}^2\right)}{\sqrt{\pi}} = 2^{-n/2} \pi^{-n(n+1)/4} \exp\left(-\frac{1}{2} \text{Tr} \hat{M}^2\right).$$

Then, we know the distribution of eigenvectors and eigenvalues is the product of the distribution of entries times the Jacobian of the diagonalization, that is

$$p(\hat{\Lambda}, \hat{O}) = p(\hat{M}) \prod_{i < j} |\lambda_j - \lambda_i|$$

Now, for the distribution on the eigenvalues, we integrate on $\mathbb{V}_n$ as

$$p(\hat{\Lambda}) = \int_{\mathbb{V}_n} p(\hat{M}) \prod_{i < j} |\lambda_j - \lambda_i| d\hat{O} = p(\hat{M}) \prod_{i < j} |\lambda_j - \lambda_i| \text{Vol}(\mathbb{V}_n)$$

which lead us to write

$$p(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{1}{Z_{n,\beta=1}} \exp\left(-\frac{1}{2} \sum_{i=1}^n \lambda_i^2\right) \prod_{i < j} |\lambda_i - \lambda_j|.$$

with $Z_{n,\beta}$ the partition function, or normalizer of the expression. We could also do the same calculation for the GUE and GSE ensembles, which would result in a similar expression

$$\begin{aligned} p(\lambda_1, \lambda_2, \dots, \lambda_N) &= \frac{1}{N! Z_{N,\beta}^{(ord)}} \exp\left(-\sum_{i=1}^N \frac{\lambda_i^2}{2} - \sum_{i < j} \log |\lambda_i - \lambda_j|^\beta\right), \\ &= \frac{1}{Z_{N,\beta}} e^{-\beta_N \mathcal{H}_N(\vec{\lambda})}, \end{aligned} \tag{1.2.4}$$

a very familiar expression for a physicist, as it is a Gibbs measure.

1.2.3 Empirical Measure

In many application one is usually concerned with the limiting behavior of the spectra as the size of the matrix tends to infinity, either to properly understand large matrices or to approximate infinite-dimensional operators. A usual tool to describe the eigenvalues of a random matrix is the empirical measure: If $\hat{M}$ has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, then its empirical spectral measure μ_M is the random probability measure putting equal mass at each of the eigenvalues:

$$\mu_M = \frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i}$$

We want to say that this measure converges, in some sense, for some other measure. A common way of doing this is the concept of *convergence in expectation*. Define $\mathbb{E}(\mu_M)$ to be as follows: for a test function f , define

$$\int f d\mathbb{E}(\mu_M) := \mathbb{E}\left(\int f d\mu_M\right)$$

If we have a sequence of matrices $\{\hat{M}_n\}$ we say that their spectral measure converges in expectation to a limiting measure μ if the sequence $\{\mathbb{E}(\mu_{M_n})\}$ converges weakly to μ , that is, if

$$\int f d\mathbb{E}(\mu_{M_n}) \xrightarrow{n \rightarrow \infty} \int f d\mu$$

for bounded continuous test functions f . A natural class of test functions are polynomials. Note that if $\lambda_1, \lambda_2, \dots, \lambda_n$ are the random eigenvalues of $\hat{M}$ random matrix, we have that

$$\int (z^k + c_{k-1}z^{k-1} + \dots + c_0) d\mu_M = \frac{1}{n} \sum_{i=1}^n (\lambda_i^k + c_{k-1}\lambda_i^{k-1} + \dots + c_0) = \frac{1}{n} \text{Tr}(V(\hat{M}))$$

In a way, the potential somehow influences the convergence.

1.3 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those self-adjoint matrices X with p.d.f.s $P(X)$ such that one has the decomposition

$$P(X) = \prod_{i=1}^n w(\lambda_i),$$

where $\{\lambda_i\}$ are the undistinguishable eigenvalues of X and $w(\lambda)$ is some continuous weight function that is real valued on $\mathbb{R}$ and decays fast enough as $|\lambda| \rightarrow \infty$. This is the same as removing the constrain on the weight function of the classical ensembles.

Definition 1.3.1. Let $\beta = 1, 2, 4$, fix $n \in \mathbb{N}$ and let $w(\lambda)$ be a continuous, non-negative, real-valued function such that $w(\lambda) = (\lambda^{-\beta(n-1)-1})$. Then, the j.p.d.f.

$$p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{N_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta$$

describes the n -sized orthogonal polynomial ensemble (OPE) when $\beta = 2$, skew-orthogonal polynomial ensemble of real type ($\mathbb{R}$ -SOPE) when $\beta = 1$, and the skew-orthogonal polynomial ensemble of quaternion type ($\mathbb{H}$ -SOPE) when $\beta = 4$.

The condition on the weight function guarantees that the j.p.d.f has a well-defined normalization constant. For the nomenclature used, let's discuss the Vandermonde determinant. Check 1.3 for this.

If one wishes to integrate the determinantal and Pfaffian structures given by the Jacobian against the product of weight functions, it is beneficial to use monic polynomials $\{q_j(\lambda)\}_{j=0}^\infty$ that are skew-orthogonal with respect to the weight $w(\lambda)$.

Definition 1.3.2. With $w(\lambda)$ a suitable function, define the following inner products on polynomial space:

$$\langle f, g \rangle^{(1)} := \frac{1}{2} \int_{\mathbb{R}^2} f(x)g(y) \operatorname{sign}(y-x) w(x)w(y) dx dy \quad (1.3.1)$$

$$\langle f, g \rangle^{(2)} := \int_{\mathbb{R}} f(x)g(x)w(x)dx \quad (1.3.2)$$

$$\langle f, g \rangle^{(4)} := \frac{1}{2} \int_{\mathbb{R}} (f(x)g'(x) - f'(x)g(x))w(x)dx \quad (1.3.3)$$

For $\beta = 1, 2, 4$, let $\{q_j^{(\beta)}\}_{j=0}^\infty$ be a family of monic polynomials such that each $q_j^{(\beta)}(\lambda)$ is of degree j . Impose also that exists finite positive constants $\{r_j^{(\beta)}\}_{j=0}^\infty$ s.t.

$$\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = r_j^{(2)} \chi_{j=k} \quad (1.3.4)$$

$$\langle q_{2j}^{(\sigma)}, q_{2k+1}^{(\sigma)} \rangle^{(\sigma)} = -\langle q_{2k+1}^{(\sigma)}, q_{2j}^{(\sigma)} \rangle^{(\sigma)} = r_j^{(\sigma)} \chi_{j=k} \quad (1.3.5)$$

$$\langle q_{2j}^{(\sigma)}, q_{2k}^{(\sigma)} \rangle^{(\sigma)} = \langle q_{2j+1}^{(\sigma)}, q_{2k+1}^{(\sigma)} \rangle^{(\sigma)} = 0 \quad (1.3.6)$$

for $\sigma = 1, 4$ and any $j, k > 0$. Then, we say that the family $\{q_j^{(1)}(\lambda)\}_{j=0}^\infty$ is $\mathbb{R}$ -skew-orthogonal, $\{q_j^{(2)}(\lambda)\}_{j=0}^\infty$ is orthogonal, and $\{q_j^{(4)}(\lambda)\}_{j=0}^\infty$ is $\mathbb{H}$ -skew-orthogonal. All in respect to $w(\lambda)$.

We can now express the eigenvalue j.p.d.f $p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta)$ as either a determinant or a Pfaffian.

Proof. Write sketch of the proof here. □

Theorem 1.3.3. Let $\{q_j^{(\beta)}(x)\}_{j=0}^\infty, \{r_j^{(\beta)}\}_{j=0}^\infty$, be as in the definition above, and let $n \in \mathbb{N}$ with n even in the case

for $\beta = 1$. Next, introduce the auxiliary functions

$$\phi_j(x) := \int_{\mathbb{R}} q_j^{(1)}(y) \operatorname{sign}(x-y) w(y) dy \quad (1.3.7)$$

$$S_j^{(1)}(x, y) := \phi'_{2j+1}(x) \phi_{2j}(y) - \phi'_{2j}(x) \phi_{2j+1}(y) \quad (1.3.8)$$

$$S_j^{(4)}(x, y) := q'_{2j+1}(x) q_{2j}(y) - q'_{2j}(x) q_{2j+1}(y) \quad (1.3.9)$$

$$D_j^{(4)}(x, y) := q'_{2j+1}(x) q'_{2j}(y) - q'_{2j}(x) q'_{2j+1}(y) \quad (1.3.10)$$

$$I_j^{(4)}(x, y) := q_{2j+1}(x) q_{2j}(y) - q_{2j}(x) q_{2j+1}(y) \quad (1.3.11)$$

and the kernels corresponding respectively to the real, complex and quaternionic cases,

$$K_n^{(1)}(x, y) := \sum_{k=0}^{N/2-1} \frac{1}{r_k^{(1)}} \begin{bmatrix} \frac{1}{2} \int_{\mathbb{R}} \operatorname{sign}(x-z) S_k^{(1)}(z, y) & S_k^{(1)}(y, x) \\ -S_k^{(1)}(x, y) & \delta_y S_k^{(1)}(x, y) \end{bmatrix} - \frac{1}{2} \begin{bmatrix} \operatorname{sign}(x-y) & 0 \\ 0 & 0 \end{bmatrix}, \quad (1.3.12)$$

$$K_n^{(2)}(x, y) := \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(2)}} q_k^{(2)}(x) q_k^{(2)}(y), \quad (1.3.13)$$

$$K_n^{(4)}(x, y) := \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(4)}} \begin{bmatrix} I_k^{(4)}(x, y) & S_k^{(4)}(y, x) \\ -S_k^{(4)}(x, y) & D_k^{(4)}(x, y) \end{bmatrix}. \quad (1.3.14)$$

Then, the eigenvalue j.p.d.f. of Definition 1.3.1 given by

$$p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \begin{cases} \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^n, & \beta = 2, \\ \operatorname{Pf} \left[K_n^{(\beta)}(\lambda_i, \lambda_j) \right]_{i,j=1}^n, & \beta = 1, 4. \end{cases}$$

The significance of this theorem is that, for $\beta = 1, 2, 4$ the kernels $K_n^{(\beta)}(x, y)$ have *Reproducing properties* such as

$$\int_{\mathbb{R}} \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^{k+1} d\lambda_{k+1} = (n-k) \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k \quad (1.3.15)$$

$$\int_{\mathbb{R}} \operatorname{Pf} \left[K_n^{(\beta)}(\lambda_i, \lambda_j) \right]_{i,j=1}^{k+1} d\lambda_{k+1} = (n-k) \operatorname{Pf} \left[K_n^{(\beta)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k \quad (1.3.16)$$

Thus, one may integrate the j.p.d.f. over $d\lambda_{k+1} \cdots d\lambda_n$ to obtain the *k-point correlation function*

$$p_k^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_k; n; \beta) := \frac{n!}{(n-k)!} \int_{\mathbb{R}^{n-k}} p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) d\lambda_{k+1} \cdots d\lambda_n = \begin{cases} \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k, & \beta = 2, \\ \operatorname{Pf} \left[K_n^{(\beta)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k, & \beta = 1, 4. \end{cases}$$

This implies, that the eigenvalue density $p^{(w)} = p_1^{(w)}(\lambda)$ is given by the formula

$$p^{(w)}(\lambda; \beta) = \begin{cases} K_n^{(2)}(\lambda, \lambda), & \beta = 2, \\ \operatorname{Pf} \left[K_n^{(\beta)}(\lambda, \lambda) \right], & \beta = 1, 4. \end{cases}$$

The case for $\beta = 2$

The formula can be further simplified. This is easier with $\beta = 2$, so that's the case we will deal with for now. First, it is known that the orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ satisfy the three-term recurrence

$$q_j^{(2)}(\lambda) = (\lambda + a_j) q_{j-1}^{(2)}(\lambda) - \frac{r_{j-1}^{(2)}}{r_{j-2}^{(2)}} q_{j-2}^{(2)}(\lambda)$$

where the coefficients depend on the choice of weight. We use this recurrence to write the terms in the summation of $K_n^{(2)}$ so that it gives rise to the *Christoffel-Darboux* formula:

Proposition 1.3.4. *The $\beta = 2$ kernel defined before simplifies as*

$$K_n^{(2)}(x, y) = \frac{\sqrt{w(x)w(y)}}{r_{n-1}^{(2)}} \frac{q_n^{(2)}(x)q_{n-1}^{(2)}(y) - q_{n-1}^{(2)}(x)q_n^{(2)}(y)}{x - y} \quad (1.3.17)$$

Taking the limit $y \rightarrow x$ using L'Hopital's rule shows that for finite n the density has the form

$$p^{(w)}(\lambda) = K_n^{(2)}(\lambda, \lambda) = \frac{\sqrt{w(x)w(y)}}{r_{n-1}^{(2)}} \left(q_{n-1}^{(2)}(\lambda)\Delta_\lambda q_n^{(2)}(\lambda) - q_n^{(2)}(\lambda)\Delta_\lambda q_{n-1}^{(2)}(\lambda) \right) \quad (1.3.18)$$

All that remains to make $p^{(w)}(\lambda)$ fully explicit is knowledge of the orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^\infty$. In the Gaussian, Laguerre and Jacobi cases, there are Hermite, Laguerre and Jacobi orthogonal polynomials, respectively.

1.4 Vandermonde

We have said before that the change of variables performed for the invariant ensembles give rise to an Jacobian of the form

$$\prod_{i < j}^n |\lambda_j - \lambda_i|^\beta,$$

lets try to properly understand this object. The first thing to note is that this is a determinant. This is done in two steps: First, we show that for a given matrix form, the determinant has exactly this form, secondly we show that the Jacobian is exactly this matrix. For the first part, consider a matrix V given by

$$V = \begin{vmatrix} 1 & \alpha_1 & \alpha_1^2 & \dots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \dots & \alpha_2^{n-1} \\ 1 & \alpha_3 & \alpha_3^2 & \dots & \alpha_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \dots & \alpha_n^{n-1} \end{vmatrix}$$

We can use of the properties of the determinant, namely that it does not change if we add a multiple of one row to another row to write

$$|V| = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & \alpha_2 - \alpha_1 & \alpha_2(\alpha_2 - \alpha_1) & \dots & \alpha_2^{n-2}(\alpha_2 - \alpha_1) \\ 1 & \alpha_3 - \alpha_1 & \alpha_3(\alpha_3 - \alpha_1) & \dots & \alpha_3^{n-2}(\alpha_3 - \alpha_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n - \alpha_1 & \alpha_n(\alpha_n - \alpha_1) & \dots & \alpha_n^{n-2}(\alpha_n - \alpha_1) \end{vmatrix} = \prod_{i=2}^n (\alpha_i - \alpha_1) \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}$$

Using now of Laplace Theorem we now that there is only one cofactor that matters. So we write

$$|V| = \prod_{i=2}^n (\alpha_i - \alpha_1) \begin{vmatrix} 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}$$

Repeating this many times we of course have the expression we started with, that is

$$|V| = \prod_{i < j}^n |\alpha_j - \alpha_i|.$$

1.4.1 Some properties

We can now use of the properties of the Vandermonde determinant to rewrite the Jacobian as we need. For reasons that will become clearer, we will use the following properties

Afirmation 1.4.1. *Let $\hat{V}$ be a matrix with entries $\{V_{i,j}\} = v_i^{j-1}$ such that its determinant is given by $\prod_{i < j}^n |v_j - v_i|$. Then, if we define $\hat{V}'$ with $\{V_{i,j}\} = p(v_i)$ with p a monomial with degree j .*

Chapter 2

Asymptotic

2.1 Asymptotic on Integrals

We will try to explore the asymptotic techniques on a specific type of function, as $\lambda \rightarrow \infty$, given by

$$\phi(\lambda) := \int_{\Gamma} g(s) e^{-\lambda f(s)} ds, \quad (2.1.1)$$

where Γ is a contour in the plane and f, g are given **suitable** functions.

2.1.1 Motivations

- Airy Equation $\psi''(x) = x\psi(x)$: Taking the Fourier transform we have

$$-\omega^2 \Phi(x) = i\Phi'(x)$$

for which we can solve for $\Phi(x)$ and then take the inverse Fourier Transform to get

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i/3\omega^3 + ix\omega} d\omega.$$

The imaginary part of the argument is an odd function and therefore zero when integrated. The rest is expressed

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos\left(\frac{\omega^3}{3} + x\omega\right) d\omega.$$

But how to know if this is the only solution? One can prove that for a more general definition of a Transform

$$\tilde{F}f(w) := \frac{1}{\sqrt{2\pi i}} \int_{\Gamma} e^{-i\omega s} f(s) ds$$

holds very similar properties of the Fourier Transformation and solves the equation for any contour that gives a convergent integral. We can actually choose independent contour such that each give us a different independent solution to the equation.

2.2 Watson Lemma

We start this study by looking at the functions $F : \mathbb{R} \rightarrow \mathbb{R}$

$$F(\lambda) = \int_0^T f(t) e^{-\lambda t} dt \quad (2.2.1)$$

which can be identified as the Laplace transform of a function f supported on $[0, T]$.

Theorem 2.2.1. (*Watson's Lemma*) Take $f \in L^1(0, T)$ and suppose that for some $\delta > 0$ it is of the form

$$f(t) = t^\alpha f_0(t), \quad 0 < t \leq \delta,$$

where $\alpha \in (-1, \infty)$ and f_0 is continuously differentiable on $[0, \delta]$ with $f_0(0) \neq 0$. Then the function F in 2.2.1 satisfies

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + \mathcal{O}(\lambda^{-1})) \quad (2.2.2)$$

when $\lambda \rightarrow \infty$ and where Γ is the Gamma function.

Proof. A weak estimative of the integral values would be given by

$$\int_0^T f(t) e^{-\lambda t} dt \leq \|f\|_{L^1}$$

where we only use the fact that $f \in L^1(0, T)$. However, we have more information on the behavior of the function in the section $0 < t < \delta$. To use this information we start by decomposing the function in two parts

$$F(\lambda) = (1) + (2) = \int_0^\delta f(t) e^{-\lambda t} dt + \int_\delta^T f(t) e^{-\lambda t} dt.$$

For (2), we can use the weak estimative, leaving us with

$$F(\lambda) \leq \int_0^\delta f(t) e^{-\lambda t} dt + \|f\|_{L^1} e^{-\delta\lambda}.$$

Now, we remind of the condition of continuous differentiability and the hypothesis on $f(t)$ to write (1) as

$$(1) = \int_0^\delta t^\alpha (f_0(0) + r(t)) e^{-\lambda t} dt = f_0(0) \int_0^\delta t^\alpha e^{-\lambda t} dt + \int_0^\delta t^\alpha r(t) e^{-\lambda t} dt = f_0(0) \cdot (1.1) + (1.2)$$

For these two integrals we can try to simplify the problem by rewriting

$$(1.1) = \int_0^\infty t^\alpha e^{-\lambda t} dt - \int_\delta^\infty t^\alpha e^{-\lambda t} dt = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - \int_\delta^\infty t^\alpha e^{-\lambda t} dt$$

$$|(1.2)| \leq \|f'\|_\infty \int_0^\delta t^{\alpha+1} e^{-\lambda t} dt = \|f'\|_\infty \left(\int_0^\infty t^{\alpha+1} e^{-\lambda t} dt - \int_\delta^\infty t^{\alpha+1} e^{-\lambda t} dt \right) = \|f'\|_\infty \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - \int_\delta^\infty t^{\alpha+1} e^{-\lambda t} dt \right).$$

We are now only missing one ingredient to get the result, the order of

$$G(\lambda, \beta) := \int_\delta^\infty t^\beta e^{-\lambda t} dt, \quad \lambda > 1, \quad \beta > -1.$$

Using Cauchy-Schwartz we can estimate

$$\begin{aligned} |G(\lambda, \beta)|^2 &\leq \left(\int_\delta^\infty e^{-\lambda t} dt \right) \left(\int_\delta^\infty t^{2\beta} e^{-\lambda t} dt \right) \\ &= \left(\frac{e^{-\lambda\delta}}{\lambda} \right) \left(\int_\delta^\infty t^{2\beta} e^{-\lambda t} dt \right) \\ &\stackrel{(u=\lambda t)}{=} \left(\frac{e^{-\lambda\delta}}{\lambda} \right) \left(\int_\delta^\infty e^{-u} u^{2\beta} \frac{1}{\lambda^{2\beta+1}} du \right) \\ &= \frac{e^{-\lambda\delta}}{\lambda^{2\beta+2}} M_\beta \end{aligned}$$

Where M_β is independent of λ . That leave us with

$$G(\lambda, \beta) = \mathcal{O}\left(\frac{e^{-\lambda\delta/2}}{\lambda^{\beta+1}}\right) = \mathcal{O}(e^{-\epsilon\lambda})$$

where $\epsilon > 0$ depends on β and δ . Finally we can rewrite

$$(1.1) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - G(\lambda, \alpha) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + \mathcal{O}(e^{-\epsilon\lambda})$$

$$|(1.2)| \leq \|f'\|_\infty \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - G(\lambda, \alpha+1) \right) = \|f'\|_\infty \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} + \mathcal{O}(e^{-\epsilon\lambda}) \right) = \mathcal{O}(\lambda^{-\alpha-2})$$

and finally

$$F(\lambda) \leq \mathcal{O}(\lambda^{-\alpha-2}) + f_0(0) \left(\frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + \mathcal{O}(e^{-\epsilon\lambda}) \right) + \|f\|_{L^1} e^{-\delta\lambda}.$$

by the dominance of the exponential error, we can finally get the result

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + \mathcal{O}(\lambda^{-1}))$$

□

In the estimative the Gamma function and the factor of λ^{-1} is the result of integrating the linear exponent of e given by $-\lambda t$. If we, however, change this exponent to be of the general type $-\lambda t^d$ we can still have a similar result on the behavior of the function.

Proposition 2.2.2. *For $T, S > 0$, define the function*

$$G(\lambda) := \int_{-S}^T g(t) e^{-\lambda t^2} dt$$

and assume $g(x)$ absolutely integrable in $[-S, T]$, of class C^1 in a fixed neighborhood of the origin. Then the function satisfies

$$G(\lambda) = \frac{g(0)\Gamma(1/2)}{\lambda^{1/2}} + \mathcal{O}\left(\frac{1}{\lambda^{3/2}}\right) \quad (2.2.3)$$

when $\lambda \rightarrow \infty$ and where Γ is the Gamma function.

Proof. The proof is a reduction of the problem to Watson's Lemma. Assume, $T > S$ and write

$$G(\lambda) = \int_{-S}^S g(t) e^{-\lambda t^2} dt + \int_S^T g(t) e^{-\lambda t^2} dt$$

The second integral we know to be of order

$$\int_S^T g(t) e^{-\lambda t^2} dt \leq \|g\|_\infty e^{-\lambda S^2},$$

exponential on λ , so we drop it. For the first integral we rewrite

$$\begin{aligned} \int_{-S}^S g(t) e^{-\lambda t^2} dt &= \int_0^S (g(t) + g(-t)) e^{-\lambda t^2} dt \\ &\stackrel{(u=t^2)}{=} \frac{1}{2} \int_0^{S^2} \left(\frac{g(\sqrt{u}) + g(-\sqrt{u})}{\sqrt{u}} \right) e^{-\lambda u} du \end{aligned}$$

where now we can use Watson's Lemma.

□

2.3 Laplace's Method

A seemly natural extension of the type of function we were dealing with are functions of the type

$$\Phi(\lambda) := \int_a^b g(t) e^{-\lambda\phi(t)} dt \quad (2.3.1)$$

for $a, b \in \mathbb{R} \cup \{-\infty, \infty\}$ and $a < b$. For now, ϕ is a real-valued function. To begin the study let's make some assumptions. The first one is that a is finite, that means that we can set $a = 0$ without loss of generality. The second assumption is that ϕ is a increasing function on $[0, b]$, so that $\phi(0) < \phi(t)$ for any t in the interval. With that we note that the coefficient $e^{-\lambda\phi(t)} / e^{-\lambda\phi(0)}$ decreases exponentially, that is, the integral is accumulated as the neighborhood of 0,

$$\Phi(\lambda) \approx \int_0^\delta g(t) e^{-\lambda\phi(t)} dt.$$

As we know $\phi(t)$ to be increasing, it's first non zero derivative must be of even order, so that when we expanded the function by Taylor,

$$\phi(t) \approx \phi(0) + \phi_{2k+1} t^{2k+1}$$

for some $k \geq 0$ and $|t| < \delta$. As ϕ_{2k+1} , the $(2k+1)$ th-derivative, must be positive we also have that, as $\lambda \rightarrow \infty$,

$$\begin{aligned} \Phi(\lambda) &\approx e^{-\lambda\phi(0)} \int_0^\delta g(t) e^{-\lambda\phi_{2k+1}(t) \cdot t^{2k+1}} dt \\ &\stackrel{(u=t(\lambda\phi_{2k+1})^{1/(2k+1)})}{\approx} e^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k+1})^{\frac{1}{2k+1}}} \int_0^\infty e^{-u^{2k+1}} du \\ &\stackrel{(s=u^{2k+1})}{\approx} e^{-\lambda\phi(0)} \frac{g(0)}{(2k+1)(\lambda\phi_{2k+1})^{\frac{1}{2k+1}}} \int_0^\infty e^{-s} ds \end{aligned}$$

Where the integral can be evaluated as an Gamma function.

More than anything we need to focus on an important assumption that was made: that ϕ has a global minima at one of the endpoints of integration. Is that were not the case, for example if $a < 0 < b$ and $\phi(t)$ had a global minima at $t = 0$ we would now have to consider the expansion

$$\phi(t) \approx \phi(0) + \phi_{2k} t^{2k}$$

which would lead us to the expression

$$\Phi(\lambda) \approx e^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k})^{\frac{1}{2k}}} \int_{-\infty}^\infty e^{-u^{2k}} du$$

which differentiates of the previously result more importantly by the limits of integration. This is closely related to physical boundaries of the problem.

Theorem 2.3.1. (*Laplace's Method, endpoint contributions*) Let Φ be as in 2.3.1, with a finite and $g \in L^1(a, b)$. Assume that ϕ has a unique global minimum at $t = a$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form

$$g(t) = (t - a)^\alpha g_0(t)$$

for $|t - a| < \delta$ and g_0 continuously differentiable in $[a, a + \delta]$ and $g_0(a) \neq 0$, whereas ϕ is twice continuously differentiable on $[a, a + \delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \frac{1}{\lambda^{\alpha+1}} \frac{g_0(a)\Gamma(\alpha+1)}{\phi'(a)^{\alpha+1}} e^{-\lambda\phi(a)} (1 + \mathcal{O}(\lambda^{-1})) \quad (2.3.2)$$

as $\lambda \rightarrow \infty$.

Proof. Start by assuming, w.l.g, that $a = 0$ and dividing the integral in two, one in the region of interest and one in the rest of the interval

$$\Phi(\lambda) = \int_0^b g(t) e^{-\lambda\phi(t)} dt = \int_0^\delta g(t) e^{-\lambda\phi(t)} dt + \int_\delta^b g(t) e^{-\lambda\phi(t)} dt = (1) + (2).$$

Knowing that ϕ has a global minima at $t = 0$ we can assume, changing δ as needed, that $\phi(t) > \phi(\delta)$ for any $t > \delta$. That give us a simple limitation for (2):

$$(2) = \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt \leq e^{-\lambda\phi(\delta)} \int_{\delta}^b g(t) dt \leq e^{-\lambda\phi(\delta)} \|g\|_{L^1[0,b]}.$$

We now get back to (1) and start by studying the difference give by $\phi(t) - \phi(0) = s$ in a similar way that we would think of a change of variable. Define the multivariate function

$$H(t, s) := \phi(t) - \phi(0) - s$$

such that $H(0, 0) = 0$ and $\partial_t H(0, 0) \neq 0$ so that we can use the Theorem of Implicit Function to say that there exists a unique bijective and continuously differentiable function $t(s) := t$ such that

$$\begin{cases} \phi(t(s)) - \phi(0) - s = 0 \\ t'(0) = \frac{\partial_s H(0,0)}{\partial_t H(0,0)} = \frac{1}{\phi'(0)} \end{cases}$$

If we set $T = t^{-1}(\delta)$ we would then have

$$\int_0^{\delta} g(s) e^{-\lambda\phi(s)} ds = \int_0^T g(t(s)) e^{-\lambda\phi(t(s))} t'(s) ds$$

for now to apply the assumption on g . We write $g(t(s)) = t^{\alpha}(s)g_0(t(s))$ and because of Taylor,

$$t(s) = st'(0)(1 + r(s)) = \frac{s}{\phi'(0)}(1 + r(s)).$$

Joining both results,

$$\begin{aligned} \int_0^T g(t(s)) e^{-\lambda\phi(t(s))} t'(s) ds &= e^{\lambda\phi(0)} \int_0^T g(t(s)) e^{-\lambda(\phi(t(s)) - \phi(0))} t'(s) ds \\ &= e^{\lambda\phi(0)} \int_0^T t^{\alpha}(s) g_0(t(s)) e^{-\lambda s} t'(s) ds \\ &= \frac{e^{\lambda\phi(0)}}{\phi'(0)^{\alpha}} \int_0^T s^{\alpha} (1 + r(s))^{\alpha} g_0(t(s)) e^{-\lambda s} t'(s) ds \end{aligned} \tag{2.3.3}$$

where, if we set $f(s) := s^{\alpha}(1 + r(s))^{\alpha}g_0(t(s))$ we can apply Watson's Lemma. $\square$

Theorem 2.3.2. (*Laplace's Method, interior contributions*) Let Φ be as in 2.3.1, with a finite and $g \in L^1(a, b)$. Assume that ϕ has a unique global minimum at $t = c \in (a, b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form

$$g(t) = (t - a)^{\alpha} g_0(t)$$

for $|t - a| < \delta$ and g_0 continuously differentiable in $[c - \delta, c + \delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c - \delta, c + \delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = e^{-\lambda\phi(c)} \left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + \mathcal{O}(\lambda^{-3/2}) \right) \tag{2.3.4}$$

as $\lambda \rightarrow \infty$.

Proof. Although the proof may be similar there is an important distinction in the solution. Suppose we wanted to try the same approach. We would first assume, w.l.g., that $c = 0$. Then, we could write

$$\Phi(\lambda) = \int_a^b g(t) e^{-\lambda\phi(t)} dt = \int_a^{-\delta} g(t) e^{-\lambda\phi(t)} dt + \int_{-\delta}^{\delta} g(t) e^{-\lambda\phi(t)} dt + \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt = (1) + (2) + (3).$$

For (1) and (3) the result is as it was before

$$\begin{cases} (1) = \int_{\delta}^a g(-t) e^{-\lambda\phi(-t)} dt \leq e^{\lambda\phi(-\delta)} \|g\|_{L^1[a,b]} \\ (3) = \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt \leq e^{\lambda\phi(\delta)} \|g\|_{L^1[a,b]} \end{cases}$$

For (2) we would try to define a function

$$H(t, s) := \phi(t) - \phi(0) - s^2.$$

However we could not apply the Implicit Function Theorem here as its derivative $\partial_t H = \phi'$ vanishes at $t = 0$. A nice fact about this indetermination is that it doesn't come from the fact that there are no solutions to $H(t, s) = 0$ but from the fact that there are two. To solve this we will use a technique called *blow up*. First, we introduce a variable $v = v(t)$ such that $vs = t$. Now, we redefine our function of interested to be

$$L(s, v) := \frac{\phi(sv) - \phi(0)}{s^2} - 1.$$

for $s \in (-\delta, \delta) - \{0\}$. We would think this function is also singular but note that, using the Taylor Expansion of ϕ we can write

$$\phi(t) = \frac{\phi''(0)}{2}t^2 + t^3R(t)$$

which leave us with

$$L(s, v) = \frac{\phi''(0)}{2}v^2 + sv^3R(sv) - 1$$

where we can indeed apply the Implicit Function Theorem. In fact, we know $L \in C^2$ and as we want to solve for $L(s, v) = 0$, we need that the v_0 we take to satisfy $L(0, v_0) = 0$, ie,

$$v_0 = \sqrt{\frac{2}{\phi''(0)}}.$$

Now, we also know that

$$\partial_v L(0, v_0) = \phi''(0)v_0 \neq 0.$$

So we get a bijective and continuously differentiable function $v = v(s)$ such that

$$\begin{cases} L(s, v(s)) = 0 \\ v(0) = v_0 \end{cases}$$

But, $L(s, v(s))$ means exactly that

$$\phi(t) - \phi(0) = s^2$$

with $t = t(s) = sv(s)$. Getting back to our equation, we have that

$$e^{\lambda\phi(0)} \int_{-\delta}^{\delta} g(t) e^{-\lambda(\phi(t)-\phi(0))} dt = e^{\lambda\phi(0)} \int_{\alpha}^{\beta} g(sv(s))(v(s) + sv'(s)) e^{-\lambda s^2} ds$$

where we can easily calculate $\alpha = -\sqrt{\phi(-\delta) - \phi(0)}$ and $\beta = \sqrt{\phi(\delta) - \phi(0)}$. We finish the prove with Proposition [2.2.2](#). $\square$

2.3.1 Example - Gamma Function

Take the function called Gamma given by the expression

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt \tag{2.3.5}$$

We are interested in what happens to the function when we take the variable x to $+\infty$. To see what happens, let's fit it into Proposition [2.3.2](#). Rewrite equation [2.3.5](#) as

$$\int_0^{\infty} e^{-t+x \log(t) - \log(t)} dt.$$

This lead us to identify ϕ as $-\log(t)$, which would give us $\phi'(t) < 0$, that is, we would have it's minimum at $t = +\infty$. However $e^{\log(t)} = t^1$ is not integrable near the origin. Something need to change about the identification. Let's introduce a variable to rescale the integral: $xs = t$, s.t.,

$$\Gamma(x) = x^x \int_0^{\infty} e^{-\lambda(s-\log(s))} e^{-s} ds.$$

Where we have used $\lambda = x - 1$. Thus we identify $\phi(s) = s - \log(s)$ and $g(s) = e^{-s}$. in fact, now ϕ has a global minima at $s = 1$ and $\phi''(1) = 1 \neq 0$. Hence

$$\begin{aligned}\Gamma(x) &= \frac{\sqrt{2\pi} x^x e^{-x}}{(x-1)^{(-1/2)}} (1 + \mathcal{O}(x^{-1})) \\ &= \sqrt{2\pi} x^{x-1/2} e^{-x} (1 + \mathcal{O}(x^{-1})) \\ &= (x - 1/2) \log(x) - x + \frac{1}{2} \log(2\pi)\end{aligned}$$

Which is called the Stirling formula.

2.4 Steepest Descent

Let's now finally explore integrals over the complex plane with complex valued functions. We retake the integral

$$\phi(\lambda) := \int_{\Gamma} g(s) e^{-\lambda f(s)} ds, \quad (2.4.1)$$

where Γ is a contour in the complex plane and f, g are given *complex-valued, analytic functions*.

Suppose that Γ is parametrized by $\gamma : [a, b] \rightarrow \mathbb{C}$ and let us decompose the functions we have into their real and imaginary parts such as

$$f(z) = u(z) + iv(z), \quad g(z) = p(z) + iq(z), \quad \gamma(t) = \alpha(t) + i\beta(t)$$

with all functions in the decomposition being real valued. Of course we now have

$$e^{-\lambda f(z)} = e^{-\lambda u(z)} e^{-i\lambda v(z)}$$

and when λ grows large, we have the exponentially growing/decaying term $e^{-\lambda u(z)}$ and a oscillating term coming from $e^{-i\lambda v(z)}$ which is hard to control. The idea to get around this fact is to use paths in which this term is constant. If we have such a case, we would write

$$\Phi(\lambda) = e^{-i\lambda v_0} \int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t)) e^{-\lambda u(\gamma(t))} dt + i e^{-i\lambda v_0} \int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t)) e^{-\lambda u(\gamma(t))} dt \quad (2.4.2)$$

where each integral can now be solved by Laplace's method.

Theorem 2.4.1. *Consider Φ as in 2.4.1 with a smooth contour Γ . Assume that the exponent $f = u + iv$ is an analytic function on a neighborhood of Γ that satisfies*

$$v(z) \equiv v_0, \quad z \in \Gamma$$

In addition, suppose that u has a unique minimum at an interior point of Γ , say on $z_0 \in \Gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of Γ and satisfies

$$\int_{\Gamma} |g(z)| |dz| < \infty.$$

Then Φ has a leading asymptotic as $\lambda \rightarrow +\infty$ given by

$$\Phi(\lambda) = e^{-\lambda f(z_0)} \left(\frac{\sqrt{2\pi} g(z_0) e^{i\theta(z_0)}}{\lambda^{1/2} \sqrt{|f''(z_0)|}} + \mathcal{O}(\lambda^{-3/2}) \right), \quad \lambda \rightarrow +\infty,$$

where $\theta(z_0)$ is the angle of the oriented tangent of Γ at z_0 .

Proof. Fix a parametrization $\gamma : [a, b] \rightarrow \mathbb{C}$ of Γ with $\gamma(c) = z_0$, $a < c < b$, and decompose Φ as before in 2.4.2. By the assumptions on the functions f, g we can apply Laplace's method on both equations, and yields, as $\lambda \rightarrow +\infty$,

$$\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t)) e^{-\lambda u(\gamma(t))} dt = \sqrt{2\pi} e^{-\lambda u(z_0)} \left(\frac{p(z_0)\alpha'(c) - q(z_0)\beta'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + \mathcal{O}(\lambda^{3/2}) \right)$$

and

$$\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t)) e^{-\lambda u(\gamma(t))} dt = \sqrt{2\pi} e^{-\lambda u(z_0)} \left(\frac{p(z_0)\beta'(c) - q(z_0)\alpha'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + \mathcal{O}(\lambda^{3/2}) \right)$$

Using that $v_0 + u(z_0) = f(z_0)$ and also that,

$$(p(z_0)\alpha'(c) - q(z_0)\beta'(c)) + i(p(z_0)\beta'(c) - q(z_0)\alpha'(c)) = g(z_0)\gamma'(c).$$

Combining the estimative for both parts we get

$$\Phi(\lambda) = \sqrt{2\pi} e^{-\lambda f(z_0)} \left(\frac{g(z_0)\gamma'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + \mathcal{O}(\lambda^{3/2}) \right)$$

and it suffices now to get an expression for the second derivative in the denominator. Knowing the imaginary part is constant we can write

$$\frac{d}{dt}u(\gamma(t)) = \frac{d}{dt}f(\gamma(t)) = \gamma'(t)f'(\gamma(t))$$

and

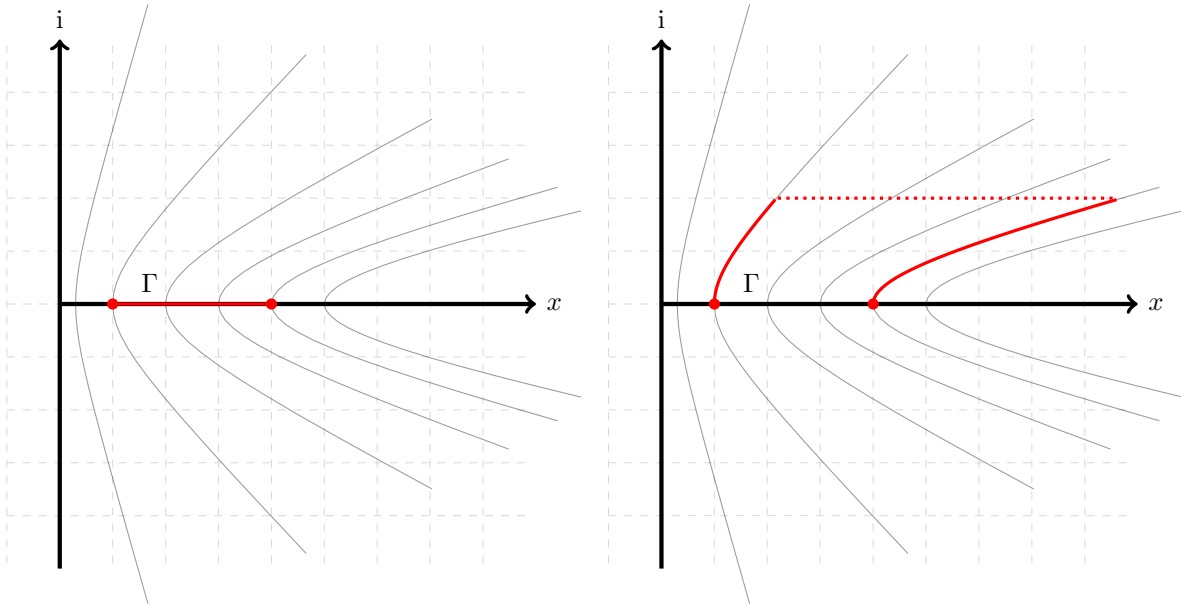
$$\frac{d^2}{dt}u(\gamma(t)) = \frac{d^2}{dt}f(\gamma(t)) = (\gamma'(t))^2 f''(\gamma(t)) + \gamma''(t)f'(\gamma(t)).$$

Hence,

$$\frac{\gamma'(c)}{\sqrt{|(u \circ \gamma)''(c)|}} = \frac{1}{\sqrt{|f''(z_0)|}} \frac{\gamma'(c)}{|\gamma'(c)|}$$

which concludes the proof. $\square$

We now need to discuss why this assumption of constant value of $\text{Im}(f(z))$ is not an assumption as strong as one might assume at first. Suppose we want to integrate in a random contour Γ of the plane. Of course, our assumption that the functions were analytic permit us to deform the contour keeping the value of the integral. We are going to use this to find a contour $\tilde{\Gamma}$ such that sections of the contour are such that $\text{Im}(f(z))$ is constant. In the figure above we plot such a scenario by using the level lines of the imaginary part of f as curves on the plane.



The name given by the method is also intuitive. We already know that being analytic means satisfying the *Cauchy-Riemann* equations,

$$u_x(z) = v_y(z), \text{ and } u_y(z) = -v_x(z).$$

An immediate result from these equations is that the gradients of u and v must be normal. Indeed, if $\langle \cdot, \cdot \rangle$ is the inner product on $\mathbb{R}^2$,

$$\langle \nabla u(z), \nabla v(z) \rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

As we're following the curves where $\text{Im}(f(z))$ is constant, we're also following the path of *steepest descent* of the real part of the integral. As we also are looking to use Laplace's method on the integrals, we may want to have our paths going through the critical points of the function.

Airy function

First, we define the function we call Airy.

$$Ai(x) := \frac{1}{2\pi i} \int_{\Gamma} e^{-\frac{1}{3}z^3 + xz} dz$$

Let's take Γ to be the union of the two half open line $\Gamma = (e^{-2\pi i/3} \infty, 0] \cup [0, e^{2\pi i/3} \infty)$. Of course, by Cauchy Theorem, we know that the definition is independent of the contour as long as the contour goes from $e^{-2\pi i/3} \infty$ to $e^{2\pi i/3} \infty$. Now, we need to get things in order to apply Laplace method on the integral. We could start defining

$$g(z) = e^{-\frac{1}{3}z^3}, \quad f(z) = z$$

but this is not a good approach as such an f won't have any critical points and won't help us calculate the integral. We then try a change of variable to get a function with those critical points. Define $x := \lambda e^{i\theta} = \lambda\omega$ so that we can send x to infinite by sending λ in a constant direction θ . Now, introduce a variable s such that $s\lambda^\eta = z$ for some $\eta \in \mathbb{R}$. Now, we want to extract λ from the new exponent

$$-\frac{1}{3}\lambda^{3\eta}s^3 + \lambda^{1+\eta}\omega s$$

so we set $\eta = 1/2$ and have that

$$Ai(\lambda, \theta) = \frac{\lambda^{\frac{1}{2}}}{2\pi i} \int_{\Gamma} e^{-\lambda^{\frac{3}{2}}(s^3/3 - \omega s)} ds$$

$$\implies Ai(\lambda, \theta) = \frac{\lambda^{\frac{1}{2}}}{2\pi i} \Phi(\lambda^{\frac{3}{2}}), \quad \Phi(\lambda) = \int_{\Gamma} e^{-\lambda f(s)} ds, \quad f(s) = f_{\omega} := \frac{s^3}{3} - \omega s$$

Note that we have f with two critical points, $s_c^{\pm} = \pm\sqrt{\omega}$. We now need to study the steepest descent path. For reasons that will be clear soon, we divide the analysis into the cases

1. $\theta = 0$
2. $0 < \theta < \frac{2}{3}\pi$
3. $\theta = \frac{2}{3}\pi$
4. $\frac{2}{3}\pi < \theta < \pi$
5. $\theta = \pi$

the rest of the cases should follow from the same arguments based on the symmetry of the critical points. For each case we want to deform Γ to a new contour $\tilde{\Gamma}$ such that

- (a) $\tilde{\Gamma}$ is homotopically equivalent to Γ , in the sense that the integral in the definition of $\Phi(\lambda)$ is invariant under the deformation from Γ to $\tilde{\Gamma}$.
- (b) The contour $\tilde{\Gamma}$ is a level line of $\text{Im}(f)$ that passes through one of the critical points of f .

For (a) to be true we need to guarantee that the deformed path $\tilde{\Gamma}$ should start at $\infty e^{-i\theta}$ and end at $\infty e^{i\theta}$ for some $\theta \in [\pi/2, 5\pi/6]$. Now we need to discuss how to satisfy (b) for every one of the cases cited. For all cases, we refer to Image 2.1 for illustration of the scenario and intuition on the contour lines we take.

For the first case, (1), we can see that the integration path can be taken as the union of the lines emerging from s_c^- with angle $\pi/2$ and $3\pi/2$. For the second case we note that something happens when $\theta = 2/3\pi$. This is because $f(s_c^+) = f(s_c^-)$ when $\theta = 2/3\pi$. But until this point the topology is equivalent. For this interval of values then we take the path of integration to be basically the same as before, the union of the lines emerging from s_c^- with angle $\pi/2$ and $3\pi/2$.

For both of these cases we have the integration of Γ to be given by the contribution of the critical point s_c^- by the asymptotic of Laplace.

For $\theta = 2/3\pi$, case (3), something change. We now need to take a path that passes through both critical points. We would think that now we have to consider the asymptotic from both saddle points but, in reality, because $\text{Re}(f(s_c^-)) < \text{Re}(f(s_c^+))$ the contribution from s_c^- exponentially dominates the asymptotic.

For case (4) we no longer have a clear connection of the lines. We need to take $\tilde{\Gamma}$ to be the union of two disjoint curves: The line coming from $\infty e^{-i2\pi/3}$ passing through s_c^- and going to infinity parallel to the real line and the line coming from infinity parallel to the real line, passing through s_c^+ and going to $\infty e^{i2\pi/3}$. Again, the contribution from s_c^- exponentially dominates the asymptotic.

For case (5), when $\theta = \pi$, the path of steepest descent is similar to the one just taken. But a key difference is to be noted, now, as $Re(f(s_c^-)) = Re(f(s_c^+))$, contributions from both saddle points must be accounted in the asymptotic.

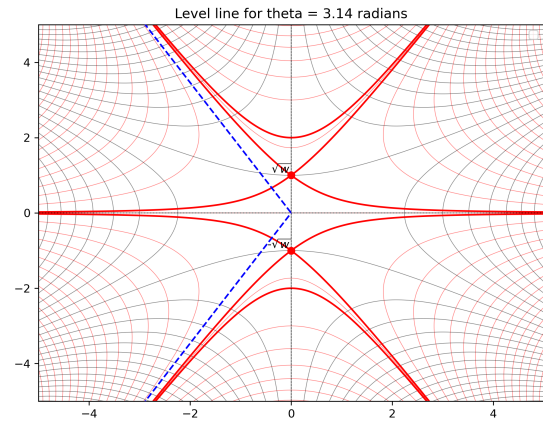
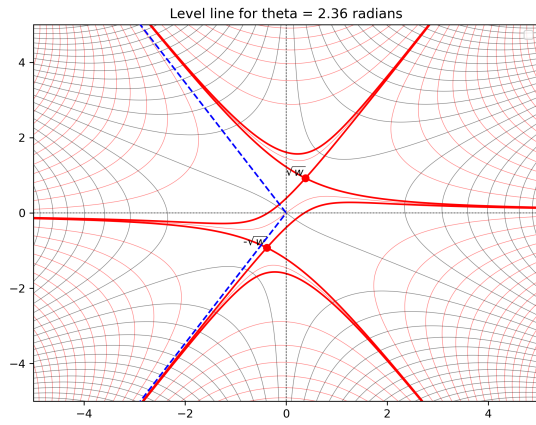
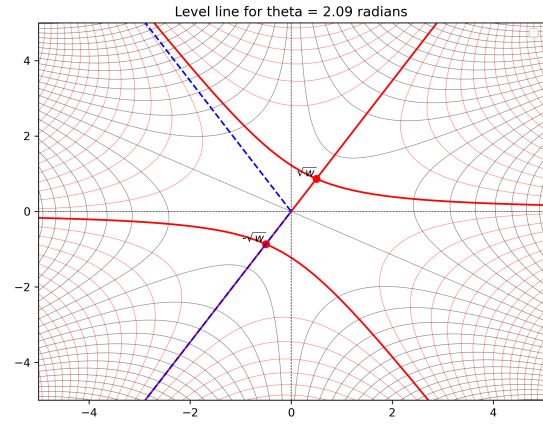
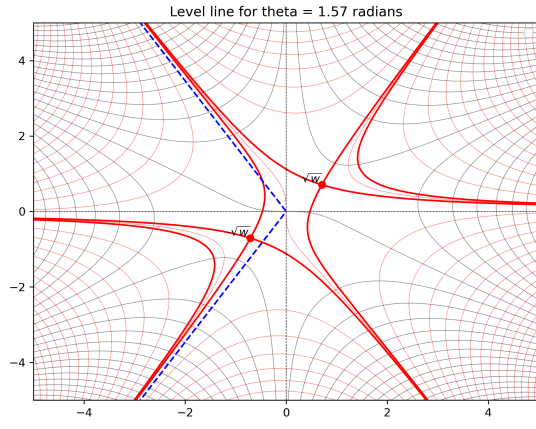
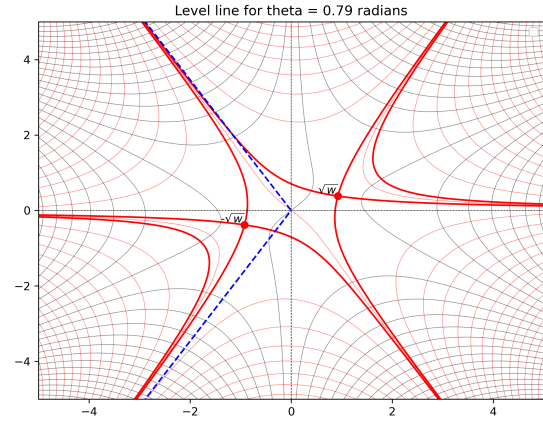
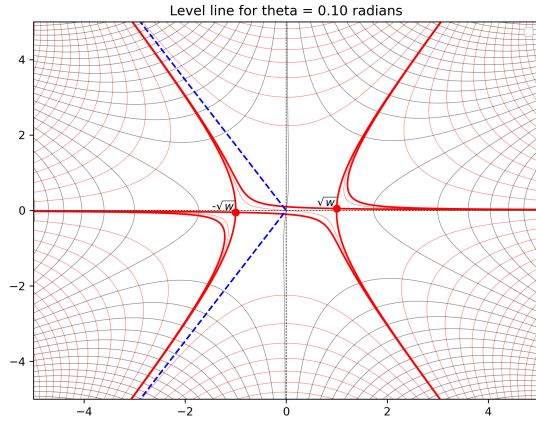


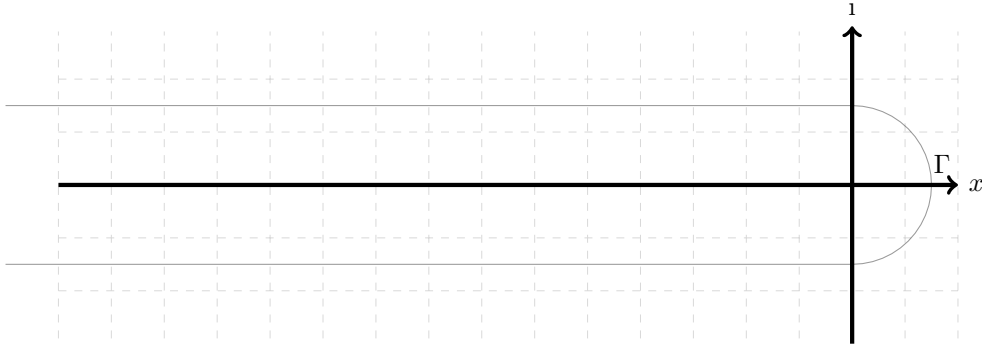
Figure 2.1: Scenarios for θ

Hermite's Polynomials

For another example, let's explore the Hermite polynomial. We first note that such polynomials have an integral representation given by

$$\begin{aligned} H_n(x) &:= \frac{n!}{2\pi i} \int_{\Gamma} z^{-n-1} e^{2xz-z^2} dz \\ &= \frac{n!}{2\pi i} \int_{\Gamma} e^{2xz-z^2-(n+1)\log(z)} dz \end{aligned} \quad (2.4.3)$$

Where Γ is a contour that goes around the origin coming and going from minus infinity such as $\Gamma = \{\lambda - i\delta : \lambda < 0\} \cup \{\delta e^{i\theta} : -\pi/2 < \theta < \pi/2\} \cup \{\lambda + i\delta : \lambda < 0\}$.



There are a few expressions we want to have. Namely we want to determine the formulas for H_n and H_{n-1} . Let's start working on the first case. To do this, we only need to consider the scaling $x \mapsto N^\alpha x$ and $z = N^\beta s$

$$\begin{aligned} H_n(x) &= \frac{n!n^b}{2\pi i} \int_{\Gamma} e^{2(n^\alpha \alpha)(n^b \beta) - (n^b \beta)^2 - (n+1)\log(N^b \beta)} d\beta \\ &= \frac{n!n^b e^{-(n+1)\log(n^b)}}{2\pi i} \int_{\Gamma} e^{2(n^{a+b} \alpha \beta) - n^{2b} \beta^2 - (n+1)\log(\beta)} d\beta \\ &\stackrel{(a=b=1/2)}{=} \frac{n! \sqrt{n} e^{-\frac{(n+1)}{2}\log(n)}}{2\pi i} \int_{\Gamma} e^{-N(-2\alpha \beta + \beta^2 + \log(\beta))} \beta^{-1} d\beta \end{aligned}$$

So that we have

$$H_n(\sqrt{n}\alpha) = C_n^0 \int_{\Gamma} e^{-N(\phi_\alpha(\beta))} \beta^{-1} d\beta$$

with $\phi_\alpha(\beta) = -2\alpha\beta + \beta^2 + \log(\beta)$. The same can be done in the second case, using now the reparametrization, that is

$$\begin{aligned} H_{n-1}(x) &= \frac{n!}{2\pi i} \int_{\Gamma} e^{2xz-z^2-(n+1)\log(z)} \\ &= \frac{n!n^b}{2\pi i} \int_{\Gamma} e^{2(n^\alpha \alpha)(n^b \beta) - (n^b \beta)^2 - (n+1)\log(N^b \beta)} d\beta \\ &= \frac{n!n^b e^{-(n+1)\log(n^b)}}{2\pi i} \int_{\Gamma} e^{2(n^{a+b} \alpha \beta) - n^{2b} \beta^2 - (n+1)\log(\beta)} d\beta \\ &\stackrel{(a=b=1/2)}{=} \frac{n! \sqrt{n} e^{-\frac{(n+1)}{2}\log(n)}}{2\pi i} \int_{\Gamma} e^{-N(-2\alpha \beta + \beta^2 + \log(\beta))} \beta^{-1} d\beta \end{aligned}$$

So that we have

$$H_{n-1}(\sqrt{n}\alpha) = C_n^1 \int_{\Gamma} e^{-N(\phi_\alpha(\beta))} \beta^{-\frac{1}{2}} d\beta$$

with, again, $\phi_\alpha(\beta) = -2\alpha\beta + \beta^2 + \log(\beta)$. As noted, for both cases, we have the same defining function $\phi_\alpha(\beta)$. As we want to calculate the asymptotic by steepest descent, we need to determine the critical points of such a function

and then the steepest descent paths passing through this points. We can easily determine the critical points as $\beta_c^\pm = \frac{\alpha \pm \sqrt{\alpha^2 - 2}}{2}$ as solutions to the quadratic formula given. Now, the behavior of the critical points will depend on the interval of the real parameter α . We divide it into the three cases: $0 < \alpha < \sqrt{2}$, $\alpha = \sqrt{2}$ and $\alpha > \sqrt{2}$.

Case 1

Here, with $0 < \alpha < \sqrt{2}$, as the plot 2.2 suggests, we have both critical points as complex values, this can be easily determined noting that if we set $0 < \alpha < \sqrt{2}$, we will have

$$\beta_c^\pm = \frac{\alpha}{2} \pm \frac{i}{2} \sqrt{2 - \alpha^2}.$$

And, if we parametrize $\alpha = \sqrt{2} \cos(\gamma)$ for $\gamma \in (0, \pi/2)$ we can rewrite

$$\beta_c^\pm = \frac{\sqrt{2}}{2} e^{\pm i\gamma}.$$

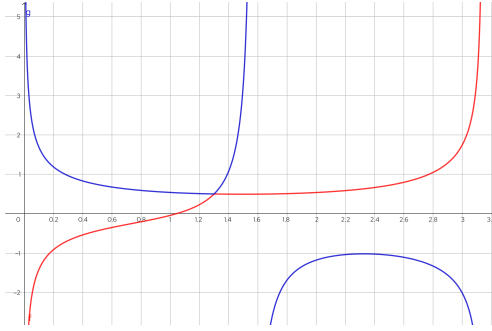
If we are indeed interested in solving the paths where the imaginary parts are constant, as needed for the Laplace method, we will have to solve an equation such as

$$\text{Im}(\phi_\alpha(z) - \phi_\alpha(\beta_c^\pm)) = 0,$$

for some $z = r e^{i\theta}$. We first solve for the positive critical point and the negative will be analogous. This can be rewritten as a quadratic equation

$$r^2 \sin(2\theta) - 2r \cos(\gamma) \sin(\theta) + (\theta - \theta_0) = 0$$

where we have set $\theta_0 = \gamma - \frac{1}{2} \sin(2\gamma)$. Where we solve for the two solutions r_a^+ and r_a^- by meshing the two quadratic solutions given by the equations. This gives us two smooth paths for the positive critical point

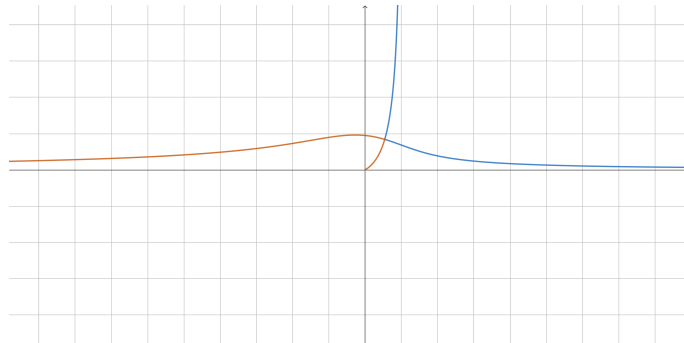


$$r_d^+(\theta) = \begin{cases} \frac{\sqrt{2} \left(\cos(\gamma) - \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/(\tan(\theta))} \right)}{2 \cos(\theta)}; & \theta \in [\gamma, \pi) \\ \frac{\sqrt{2} \left(\cos(\gamma) + \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/(\tan(\theta))} \right)}{2 \cos(\theta)}; & \theta \in (0, \gamma) \end{cases}$$

$$r_a^+(\theta) = \begin{cases} \frac{\sqrt{2} \left(\cos(\gamma) + \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/(\tan(\theta))} \right)}{2 \cos(\theta)}; & \theta \in [\gamma, \pi/2) \\ \frac{\sqrt{2} \left(\cos(\gamma) - \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/(\tan(\theta))} \right)}{2 \cos(\theta)}; & \theta \in [\theta_0, \gamma) \end{cases}$$

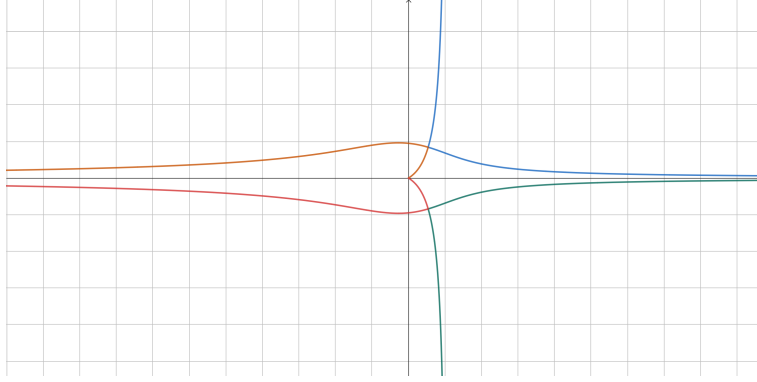
With which we can construct both paths of constant imaginary part

$$\Gamma_d^+ = \{r_d^+(\theta) e^{i\theta} : \theta \in (0, \pi)\}; \quad \Gamma_a^+ = \{r_a^+(\theta) e^{i\theta} : \theta \in (0, \pi/2)\}$$



Analogously, for the negative part,

$$\Gamma_d^- = \{r_d^-(\theta) e^{i\theta} : \theta \in (0, \pi)\}; \quad \Gamma_a^- = \{r_a^-(\theta) e^{i\theta} : \theta \in (0, \pi/2)\}$$



Now, we only need to determine the steepest descent path. This should be easy as we know the asymptotic behaviour $\phi_\alpha(t) \asymp t^2$ for $t \rightarrow \infty$ and that $\phi_\alpha(t) \asymp \ln(t)$ for $t \rightarrow 0$. From this, follows that Γ_d^+ and Γ_d^- are the paths we were looking for. We create therefore, a new path Γ_x that connects both Γ_d at the vertical line x . Then, doing $\Gamma \mapsto \Gamma_d^+ \cup \Gamma_d^- \cup \Gamma_x$ and noting that, as x goes to infinity, we have

$$\left| \int_{\Gamma_x} g(x) e^{-n\phi_\alpha(\beta)} d\beta \right| \leq \max_{y \in \Gamma_x} |g(x) e^{-n\phi_\alpha(\beta)}| \int_{\Gamma_x} |d\beta| \xrightarrow{t \rightarrow \infty} 0$$

where $g(t)$ can be both the inverse square root or inverse function. Therefore, we only need to consider the other two paths when considering the asymptotic.

Finally, retaking the expression for the Hermite Polynomial,

$$\begin{aligned} H_n(\sqrt{2n} \cos(\gamma)) &= C_n^0 \int_{\Gamma} e^{-N(\phi_\alpha(\beta))} \beta^{-1} d\beta = C_n^0 \left[\int_{\Gamma_d^-} e^{-N(\phi_\alpha(\beta))} \beta^{-1} d\beta + \int_{\Gamma_d^+} e^{-N(\phi_\alpha(\beta))} \beta^{-1} d\beta \right] \\ &\stackrel{(Steepest)}{=} C_n^0 \frac{\sqrt{2\pi}}{\sqrt{2n}} \left[\frac{e^{-n(\phi_\alpha(\beta_c^+))} e^{-i(\theta(\beta_c^+))}}{e^{i\gamma/2}/2} \frac{1}{4 \sin(\gamma)} + \frac{e^{-n(\overline{\phi_\alpha(\beta_c^+)})} e^{-i(\overline{\theta(\beta_c^+)})}}{e^{-i\gamma/2}/2} \frac{1}{4 \sin(\gamma)} \right] \\ &= C_n^0 \frac{\sqrt{2\pi}}{2 \sin(\gamma) \sqrt{2n}} \left[e^{-i(\gamma/2 + \theta(\beta_c^+)) - n\phi_\alpha(\beta_c^+)} + e^{i\pi} e^{i(\gamma/2 + \theta(\beta_c^+)) - n\overline{\phi_\alpha(\beta_c^+)}} \right] \\ &= C_n^0 \frac{\sqrt{2\pi}}{2 \sin(\gamma) \sqrt{2n}} \left[e^{-i(\gamma/2 + \theta(\beta_c^+)) - n(-2\alpha\beta_c^+ + (\beta_c^+)^2 + \log(\beta_c^+))} + e^{i\pi} e^{i(\gamma/2 + \theta(\beta_c^+)) - n(-2\alpha\overline{\beta_c^+} + (\overline{\beta_c^+})^2 + \log(\overline{\beta_c^+}))} \right] \\ &= C_n^0 \frac{\sqrt{\pi} e^{-n \ln \frac{\sqrt{2}}{2}}}{2 \sin(\gamma) \sqrt{n}} \left[e^{-i(\gamma/2 + \theta(\beta_c^+) + n\gamma) - n e^{i\gamma}(-\alpha\sqrt{2} + \frac{e^{i\gamma}}{2})} - e^{i(\gamma/2 + \theta(\beta_c^+) + n\gamma) - n e^{-i\gamma}(-\alpha\sqrt{2} + \frac{e^{-i\gamma}}{2})} \right] \end{aligned}$$

Now, since

$$\begin{cases} \operatorname{Re}[-n e^{i\gamma}(-\alpha\sqrt{2} + \frac{e^{i\gamma}}{2})] = \frac{-\sqrt{2}}{2}n; & \operatorname{Im}[-n e^{i\gamma}(-\alpha\sqrt{2} + \frac{e^{i\gamma}}{2})] = 0 \\ \operatorname{Re}[-n e^{-i\gamma}(-\alpha\sqrt{2} + \frac{e^{-i\gamma}}{2})] = \frac{-\sqrt{2}}{2}n; & \operatorname{Im}[-n e^{-i\gamma}(-\alpha\sqrt{2} + \frac{e^{-i\gamma}}{2})] = 0. \end{cases}$$

We will have that,

$$\begin{aligned} H_n(\sqrt{2n} \cos(\gamma)) &= C_n^0 \frac{\sqrt{\pi} e^{-n \ln \frac{\sqrt{2}}{2}}}{2 \sin(\gamma) \sqrt{n}} e^{-n \frac{\sqrt{2}}{2}} \left[e^{-i(\gamma/2 + \theta(\beta_c^+) + n\gamma)} - e^{i(\gamma/2 + \theta(\beta_c^+) + n\gamma)} \right] \\ &= \frac{n! \sqrt{n} e^{-\frac{(n+1)}{2} \ln(n)}}{2\pi i} \frac{\sqrt{\pi} e^{-n \ln \frac{\sqrt{2}}{2}}}{2 \sin(\gamma) \sqrt{n}} e^{-n \frac{\sqrt{2}}{2}} 2i \sin(\gamma/2 + \theta(\beta_c^+) + n\gamma) \\ &= \frac{n! n^{-\frac{(n+1)}{2}}}{2 \sin(\gamma) \pi^{\frac{3}{2}}} e^{-n(\ln \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2})} \sin(\gamma/2 + \theta(\beta_c^+) + n\gamma) \end{aligned}$$

Of course, if we change for the other case we can simply rewrite

$$\begin{aligned}
H_{n-1}(\sqrt{2n} \cos(\gamma)) &= C_n^1 \frac{\sqrt{\pi} e^{-n \ln \frac{\sqrt{2}}{2}}}{2 \sin(\gamma) \sqrt{n}} e^{-n \frac{\sqrt{2}}{2}} \left[e^{-i(\gamma + \theta(\beta_c^+) + n\gamma)} - e^{i(\gamma + \theta(\beta_c^+) + n\gamma)} \right] \\
&= \frac{n! \sqrt{n} e^{-\frac{(n+1)}{2} \ln(n)}}{2\pi i} \frac{\sqrt{\pi} e^{-n \ln \frac{\sqrt{2}}{2}}}{2 \sin(\gamma) \sqrt{n}} e^{-n \frac{\sqrt{2}}{2}} 2i \sin(\gamma + \theta(\beta_c^+) + n\gamma) \\
&= \frac{n! n^{-\frac{(n+1)}{2}}}{2 \sin(\gamma) \pi^{\frac{3}{2}}} e^{-n \left(\ln \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right)} \sin(\gamma + \theta(\beta_c^+) + n\gamma)
\end{aligned}$$

Case 2

Here, with $\alpha > \sqrt{2}$, as the plot 2.3 suggests, we have both critical points as real values, this can be easily determined noting that if we set $\alpha > \sqrt{2}$, we will have

$$\beta_c^\pm = \frac{\alpha}{2} \pm \frac{1}{2} \sqrt{\alpha^2 - 2}.$$

And, if we parametrize $\alpha = \sqrt{2} \cosh(\gamma)$ for $\gamma > 0$ we can rewrite

$$\beta_c^\pm = \frac{\sqrt{2}}{2} e^{\pm \gamma}.$$

If we are indeed interested in solving the paths where the imaginary parts are constant, as needed for the Laplace method, we will have to solve an equation such as

$$\text{Im}(\phi_\alpha(z) - \phi_\alpha(\beta_c^\pm)) = 0,$$

for some $z = r e^{i\theta}$. We first solve for the positive critical point and the negative will be analogous. This can be rewritten as a quadratic equation

$$r^2 \sin(2\theta) - 2r \cosh(\gamma) \sin(\theta) + \frac{\theta}{2} = 0$$

where we solve for the two solutions r_a^+ and r_d^- by meshing the two quadratic solutions given by the equations. This gives us two smooth paths for the positive critical point

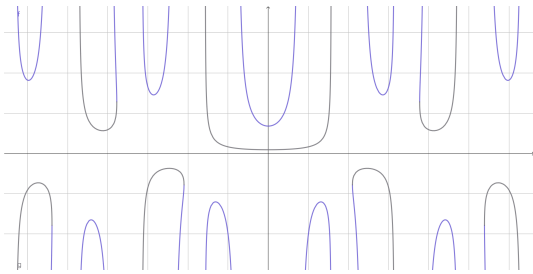
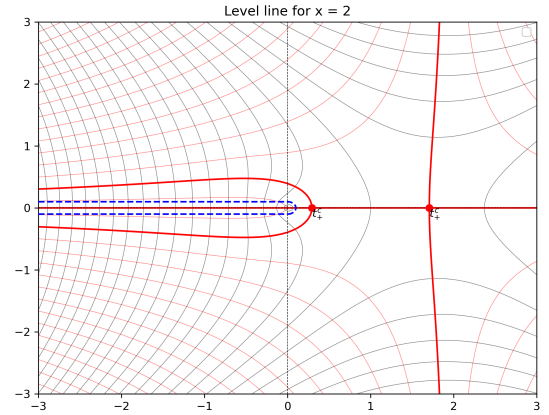


Figure 2.3: Figure for case 3

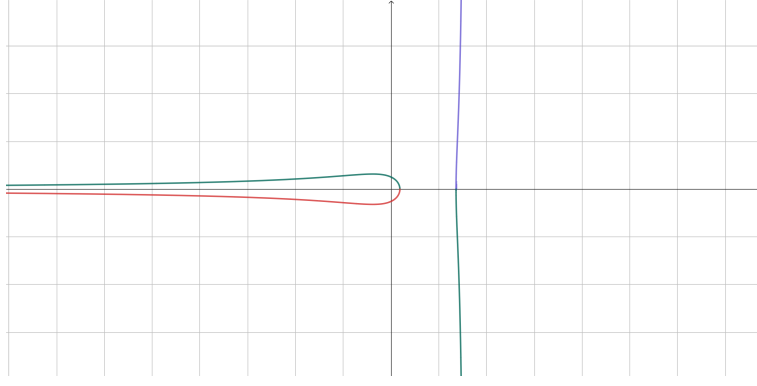


$$\begin{aligned}
r_d(\theta) &= \frac{\sqrt{2} \left(\cosh(\gamma) - \sqrt{\cosh^2(\gamma) - \frac{\theta}{\tan(\theta)}} \right)}{2 \cos(\theta)}; \quad \theta \in [0, \pi) \\
r_a(\theta) &= \frac{\sqrt{2} \left(\cosh(\gamma) + \sqrt{\cosh^2(\gamma) - \frac{\theta}{\tan(\theta)}} \right)}{2 \cos(\theta)}; \quad \theta \in [0, \pi)
\end{aligned}$$

With which we can construct both paths of constant imaginary part

$$\Gamma_d = \{r_d(\theta) e^{i\theta} : \theta \in (0, \pi)\}; \quad \Gamma_a = \{r_a(\theta) e^{i\theta} : \theta \in (0, \pi)\}$$

We need now to determine the Steepest Descent Path. In this case is more clear what the path should be the one passing through the critical point $e^{-\gamma}$. Then, repeating the process we did on Case 1 - changing the path of



integration - we can make the substitution

$$\begin{aligned}
H_{n-1}(\sqrt{2n} \cosh(\gamma)) &= C_n^1 \int_{\Gamma} e^{-N(\phi_{\alpha}(\beta))} \beta^{-\frac{1}{2}} d\beta \\
&= C_n^1 \int_{\Gamma_d} e^{-N(\phi_{\alpha}(\beta))} \beta^{-\frac{1}{2}} d\beta \\
&\stackrel{(Steepest)}{=} C_n^1 \frac{\sqrt{2\pi}}{\sqrt{2n}} \left[\frac{e^{-n(\phi_{\alpha}(\beta_c^+))}}{e^{\gamma/2}/2} \frac{e^{-i(\theta(\beta_c^+))}}{\sqrt{|1-e^{2\gamma}|}} \right] \\
&= \frac{n!n^{-\frac{(n+1)}{2}}}{i\pi^{\frac{3}{2}}} \left[\frac{e^{-n(\phi_{\alpha}(\beta_c^+))}}{e^{\gamma/2}} \frac{e^{-i(\theta(\beta_c^+))}}{\sqrt{|1-e^{2\gamma}|}} \right] \\
&= \frac{n!n^{-\frac{(n+1)}{2}}}{\pi^{\frac{3}{2}}\sqrt{|1-e^{2\gamma}|}} e^{-n(\phi_{\alpha}(\beta_c^+)+\frac{\gamma}{2n})}
\end{aligned}$$

Similary, for the other case

$$\begin{aligned}
H_n(\sqrt{2n} \cosh(\gamma)) &= C_n^0 \int_{\Gamma} e^{-N(\phi_{\alpha}(\beta))} \beta^{-1} d\beta \\
&= C_n^0 \int_{\Gamma_d} e^{-N(\phi_{\alpha}(\beta))} \beta^{-1} d\beta \\
&\stackrel{(Steepest)}{=} C_n^1 \frac{\sqrt{2\pi}}{\sqrt{2n}} \left[\frac{e^{-n(\phi_{\alpha}(\beta_c^+))}}{e^{\gamma}/2} \frac{e^{-i(\theta(\beta_c^+))}}{\sqrt{|1-e^{2\gamma}|}} \right] \\
&= \frac{n!n^{-\frac{(n+1)}{2}}}{\pi^{\frac{3}{2}}i} \left[\frac{e^{-n(\phi_{\alpha}(\beta_c^+))}}{e^{\gamma}} \frac{e^{-i(\theta(\beta_c^+))}}{\sqrt{|1-e^{2\gamma}|}} \right] \\
&= \frac{n!n^{-\frac{(n+1)}{2}}}{\pi^{\frac{3}{2}}\sqrt{|1-e^{2\gamma}|}} e^{-n(\phi_{\alpha}(\beta_c^+)+\frac{\gamma}{n})}
\end{aligned}$$

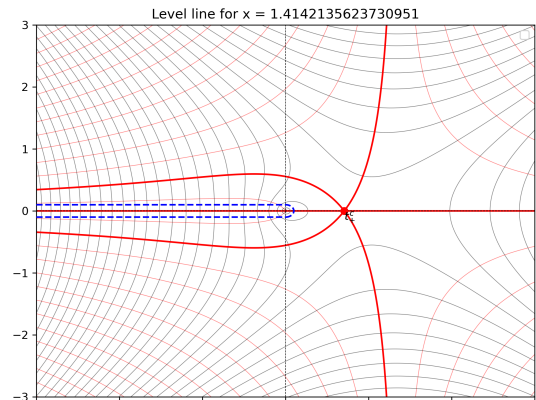
Case 3

For the case (1) we can see that the contribution for the asymptotic will come from t_c^+ while for case (2) the contribution comes from both critical points, although depending on the real part one of the asymptotic may be exponential bigger than the other. The interesting case comes from (3) where both critical point have the same value and $f'''(t_c) = 0$. We need to rescale the variable somehow to get the behavior of the asymptotic.

The idea is to perform a blow-up on the integral. We would have

$$\int_{\Gamma} e^{-N(-2xt+t^2+\log(t))} t^{-1} dt := \int_{\Gamma} e^{-N\Phi(t)} g(t) dt$$

Figure 2.4: Figure for case 3



where we know Φ to have first and second derivative equal to zero at $t_c = \sqrt{2}/2$. First we need to divide the integral in three parts

$$\int_{\Gamma_1} e^{-N\Phi(t)} g(t) dt + \int_{t_c-\delta}^{t_c+\delta} e^{-N\Phi(t)} g(t) dt + \int_{\Gamma_2} e^{-N\Phi(t)} g(t) dt \approx \int_{t_c-\delta}^{t_c+\delta} e^{-N\Phi(t)} g(t) dt$$

So we would write

$$\Phi(t) = \frac{\Phi^{(3)}(t_c)}{3!} t^3 + \mathcal{O}(t^4).$$

We now want to find a change of variable $\Phi(t) = s^3$ such that $s^3 = \Phi(t) = \tilde{\Phi}(t)t^2$. If we have such an equality we know $\tilde{\Phi}$ to be analytic, $\tilde{\Phi}'(t_c) \neq 0$ and $\tilde{\Phi}(t) = \frac{\Phi^{(3)}(t_c)}{3!} t + \mathcal{O}(t^2) := c_0 t + \mathcal{O}(t^2)$.

To find such a change of variable we need to investigate the function

$$H(t, s) = t^2 \tilde{\Phi}(t) - s^3.$$

We first note that we can not find apply the Inverse Function Theorem in such a function because it's derivative is zero at $(t_c, 0)$. For that we introduce a new variable function $v = v(t)$ such that $t = vs$. Now,

$$H(t, s) = s^3 \left(\frac{v^2}{s} \tilde{\Phi}(vs) - 1 \right) := s^3 Q(v, s).$$

We can of course also write $\partial_s Q(v, s) = v' \partial_v Q + \partial_s Q = 0$ giving us

$$\frac{\partial v(t)}{\partial s} = - \frac{\partial_s Q}{\partial_v Q}.$$

It remains to check that we can indeed apply the Implicit Function Theorem to this function Q . For that we compute the derivative of the function. As

$$Q(s, v(s)) = -1 + \frac{v^2}{s} (c_0 vs + \mathcal{O}(v^2 s^2)) = -1 + c_0 v^3 + \mathcal{O}(v^4 s)$$

and because of that

$$\partial_v Q = 3c_0 v^2 + \mathcal{O}(v^3 s) \neq 0, \quad \text{if } v(t_c) \neq 0.$$

Because of the theorem we can say that there exists a function $v(t)$ such that $H(vs, s) = 0$ and $v(t_c) = v_0$. Getting back to the integral we use $t = vs$ to write

$$\begin{aligned} &= \int_{\alpha}^{\beta} e^{-Ns^3} (v(s) + v'(s)s) g(v(s)s) ds, \quad \alpha = -\sqrt[3]{\Phi(\delta - t_c)} \quad \text{and} \quad \beta = \sqrt[3]{\Phi(\delta + t_c)} \\ &= \int_{\alpha}^{\beta} e^{-Ns^3} \frac{(v(s) + v'(s)s)}{v(s)s} ds \\ &= \int_{\alpha}^{\beta} e^{-Ns^3} \left(\frac{1}{s} + \frac{v'(s)}{v(s)} \right) ds \end{aligned} \tag{2.4.4}$$

where hopefully I can apply Laplace.

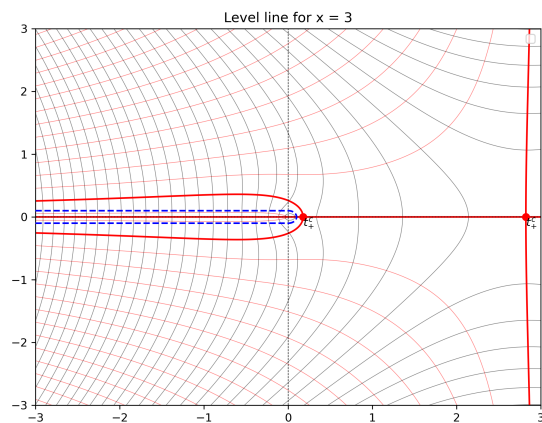
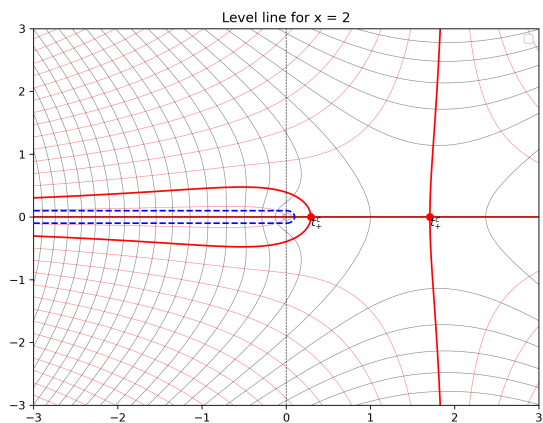
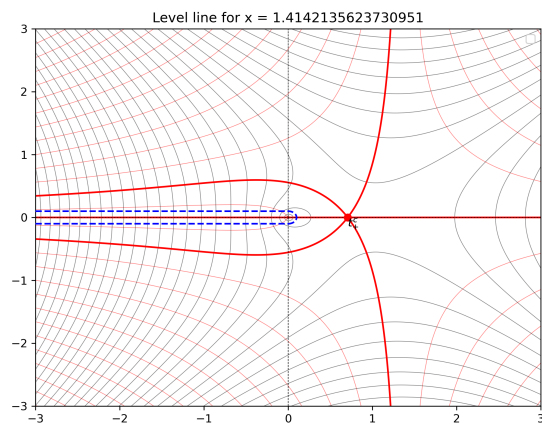
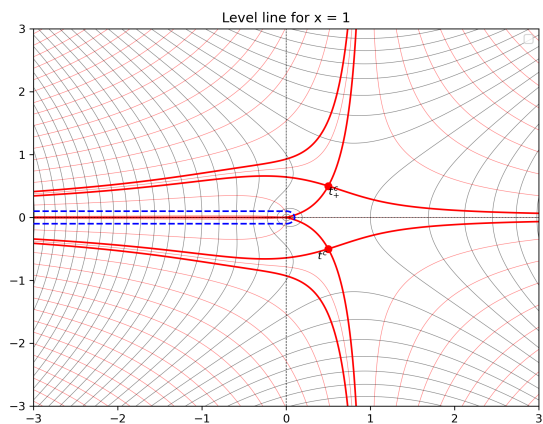
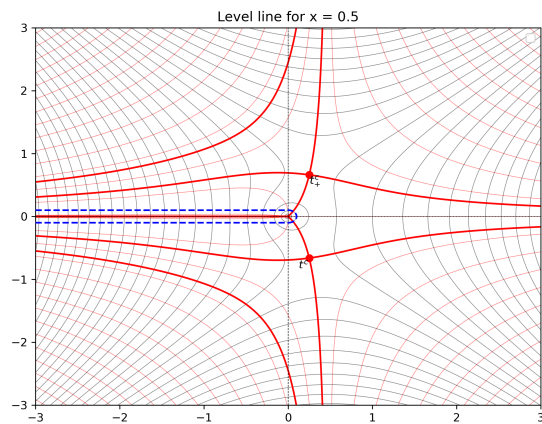
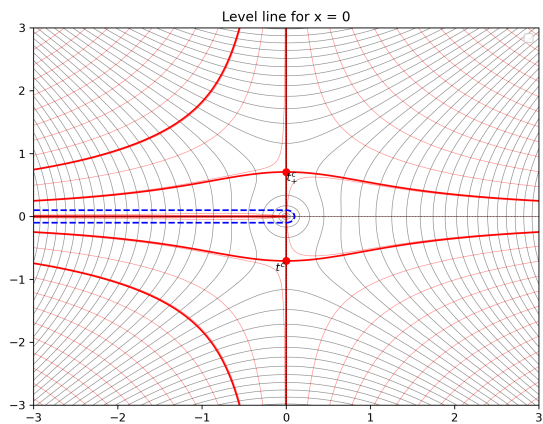


Figure 2.5: Scenarios for x

Chapter 3

Miscellaneous

3.1 Laplace Transform

3.2 Fourier Transform

We start by defining what we call a Fourier Series. Take $f(x)$ to be a a -periodic function, i.e., $f(x+a) = f(x)$, $\forall x \in \mathbb{R}$. How to characterize such a function? We may try to specify the values it takes on the interval but that's uncountable big. We can also try to approximate it somehow, one such approach is to use geometric periodic functions. We write:

$$f(x) = \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{2\pi nx}{a}\right) + \sum_{m=1}^{\infty} \beta_m \cos\left(\frac{2\pi mx}{a}\right) = \sum_{-\infty}^{\infty} f_n e^{2\pi i n \frac{x}{a}}$$

Note that such a summation must be a -periodic as all functions used are a -periodic. Of course, not all functions are guarantee to be expressed in such a way. A know class that works, and usually enough for our purposes, are square-integrable functions, for which

$$\int_{-a/2}^{a/2} |f(x)|^2 dx$$

exists and is finite. The concept we have is useful but limited if used as such. We want to extend such a concept for functions that are not a -periodic - but how? Define a family of functions $\{f_a(x) | a \in \mathbb{R}^+\}$ such that $f_a(x)$ is a -periodic and $\lim_{a \rightarrow \infty} f_a(x) = f(x)$. We have that every function f_a in the family can be written as a Fourier Series as it is periodic, so we write:

$$f_a(x) = \sum_{-\infty}^{\infty} f_{an} e^{ik_n x} : k_n = n\Delta k; \Delta k = \frac{2\pi}{a}.$$

And as we know about the limit, we can write

$$\begin{aligned} f(x) &= \lim_{a \rightarrow \infty} \left[\sum_{-\infty}^{\infty} f_{an} e^{ik_n x} \right] = \lim_{a \rightarrow \infty} \left[\sum_{-\infty}^{\infty} \frac{\Delta k}{2\pi} e^{ik_n x} \left(\frac{2\pi f_{an}}{\Delta k} \right) \right] \\ &= \int_{-\infty}^{\infty} \frac{dk}{2\pi} e^{ikx} F(k); \text{ where } F(k) = \lim_{a \rightarrow \infty} \left[\frac{2\pi f_{an}}{\Delta k} \right]. \end{aligned} \quad (3.2.1)$$

where we call $F(x)$ the Fourier Tranform of the function $f(x)$. We can also define the inverse transform

$$\begin{aligned} F(k) &= \lim_{a \rightarrow \infty} \left[\frac{2\pi}{\Delta k} f_{an}(x) \right] \\ &= \lim_{a \rightarrow \infty} \left[\frac{2\pi}{\frac{2\pi}{a}} \frac{1}{a} \int_{-a/2}^{a/2} dx e^{-ik_n x} \right] \\ &= \int_{-\infty}^{\infty} dx e^{-ikx} f(x) \end{aligned} \quad (3.2.2)$$

With this definition we can make explicit some properties of the transform, namely:

1. Linearity: Given $f(x)$ and $g(x)$, with its respective $F(x)$ and $G(x)$ Fourier Tranforms we have that

$$af(x) + bg(x) \rightarrow_{FT} aF(k) + bG(k);$$

2. Translation: Given $f(x)$ translated by b we write

$$f(x+b) \rightarrow_{FT} e^{ikb} F(k);$$

3. Derivative: Given $f(x)$ we have that the derivative is written

$$\frac{d}{dx} f(x) \rightarrow_{FT} ikF(k)$$

3.2.1 On Differential Equations

Take the damped harmonic oscillator subjected to an additional force $f(t)$. The equation of motion is given by

$$\frac{d^2x(t)}{dt} + 2\gamma\frac{dx(t)}{dt} + \omega_0^2x(t) = \frac{f(t)}{m}.$$

To solve for $x(t)$ we first take the Fourier Transform

$$-\omega^2X(\omega) - 2i\gamma\omega X(\omega) + \omega_0^2X(\omega) = \frac{F(\omega)}{m}$$

where now we can reorganize to solve for $X(\omega)$.

$$X(\omega) = \frac{\frac{F(\omega)}{m}}{-\omega^2 - 2i\gamma\omega + \omega_0^2}.$$

Now it is a simple case of taking the inverse of the Fourier Transform

$$x(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega t} \frac{F(\omega)}{m}}{-\omega^2 - 2i\gamma\omega + \omega_0^2}; \quad \text{where } F(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} f(t)$$

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